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UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP2523: DATABASE

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DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN

FOR <Hasta Vehicle Rental Booking System>

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Stakeholder: Hasta Travel & Tours Sdn Bhd

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1.0 Introduction

1.1 Overview about the company

Hasta Travel & Tours Sdn. Bhd. is a car rental company that operates as a car rental service provider, offering short-term vehicle rental service specifically for Universiti Teknologi Malaysia (UTM) students and staff. As a car rental operator, its daily operation relies heavily on accurate vehicle availability tracking, precise rental scheduling, timely payment verification and well-maintained customer records.

In addition to handling daily bookings, Hasta Travel manages a high volume of transactional data, including customer profiles, vehicle details, rental schedules, payment records, deposits, penalties, and loyalty rewards. These data elements are highly interrelated and require consistent updates throughout the rental lifecycle.

However, the current rental operation relies on a largely manual and semi-digital booking system, supported by basic tools such as WhatsApp, WhatsForm, Google Sheets , physical documents, whiteboards and the Drive Wahdah system. The company adopts a file-based system approach, where information is stored in separate files on computers rather than within a centralized database. These tools are used independently rather than as part of an integrated platform, resulting in fragmented workflows and duplicated data handling. The Drive Wahdah system functions mainly as a client-side rental record system, accessible only by staff and administrators, while customers interact with the company primarily through manual communication channels.

As a result, as the volume and complexity of rental-related data increase over time, an organized and structured data management approach becomes essential to support efficient operations and long-term scalability.

1.2 Problems

Currently, Hasta Travel does not operate a centralized database system to manage its car rental operations. Instead, operational data is distributed across multiple independent platforms such as booking requests and confirmations are exchanged through WhatsApp messages, payment records are manually tracked using Google Sheets, and customer identity details are verified through shared images and chat attachments. In addition, the Drive Wahdah system is used only as a client-side rental record platform, only accessible by staff and administrators, and does not function as an integrated data repository.

This fragmented data storage approach leads to several operational inefficiencies, including difficulty in retrieving complete booking histories, heavy reliance on staff memory and manual searching, and inconsistent records across different platforms. As data is duplicated and updated separately in WhatsApp, spreadsheets, and Drive Wahdah, synchronization cannot be guaranteed. Consequently, staff are unable to rely on a single authoritative dataset to support daily operations and decision-making.

Vehicle availability is currently not controlled through database-level mechanisms. Although booking details are recorded in spreadsheets and partially documented in the Drive Wahdah system, these platforms are not synchronized in real time. As a result, multiple bookings may be confirmed for the same vehicle within overlapping time periods. There is no automated validation at the data level to prevent scheduling conflicts, and staff must manually cross-check booking times across WhatsApp messages, spreadsheets, whiteboards, and Drive Wahdah records. From a database design standpoint, these issues arise due to the absence of a centralized Booking entity with enforced relationships to the Vehicle entity, as well as the lack of time-based integrity rules to ensure that a vehicle can only be booked once for a specific time slot. Consequently, this leads to frequent double bookings, customer dissatisfaction, and operational confusion.

Payment verification is currently handled through manual processes, where customers submit screenshots of payment receipts via WhatsApp and staff manually update payment status in Google Sheets and Drive Wahdah. However, payment information is not stored in a structured and relational manner. Receipts may be lost within chat histories, and there is no consistent linkage between booking records and payment records across platforms. Without a

dedicated Payment entity and a complete transaction history, payment status cannot be reliably queried or audited. The Drive Wahdah system records payment status at a basic level but does not maintain a comprehensive audit trail that includes receipt evidence, verification timestamps, and staff accountability. This limitation increases financial risk and makes payment dispute resolution difficult.

Blacklist management is currently performed using informal notes, spreadsheets, or staff memory rather than a structured data model. Although booking and customer information exists across spreadsheets and the Drive Wahdah system, outstanding balances and penalty records are not systematically linked to user accounts. As a result, access restrictions cannot be enforced automatically by the system. This situation allows blacklisted users to continue making bookings again, while staff must manually verify customer history for each request by checking multiple sources. This manual approach increases the risk of human error. A database-driven blacklist mechanism should be rule-based, queryable, and directly linked to user, booking, and payment records.

Manual data entry across multiple platforms, including WhatsApp, Google Sheets, whiteboards, and Drive Wahdah, results in frequent data quality issues. These include inconsistent date and time formats, missing or duplicated customer records, and incorrect booking durations or vehicle assignments. As data is repeatedly re-entered into different systems, the risk of inconsistency increases significantly. Without database-level constraints such as NOT NULL, UNIQUE, PRIMARY KEY, and FOREIGN KEY, data integrity cannot be enforced systematically. Over time, these weaknesses lead to unreliable records, reduced confidence in stored data, and increased difficulty in maintaining accurate operational information.

Although booking and payment data exists in spreadsheets and the Drive Wahdah system, the current data structure does not support efficient analytical reporting. Queries such as identifying peak booking periods, analyzing the most frequently rented vehicles, detecting users with repeated late payments, or monitoring monthly revenue trends cannot be generated easily or accurately. Because data is scattered across multiple unsynchronized platforms, reporting requires extensive manual consolidation and verification. This process is time-consuming and prone to error, significantly limiting management's ability to perform data-driven decision-making and long-term operational planning.

1.3 Proposal

Our proposed system is a usable and easy to maintain system which is implemented and designed to manage the car rental operations of the company. There are three types of system users that are categorized into customer, staff and admin. In this Hasta Car Rental system, there are several modules including user management, fleet management, booking, rewards and reporting & analytics modules.

The system can be accessed through the general website via any search engine. Users can either log into their existing account or sign up for a new account. As for staff and administrators, they are only permitted to log into the system. Staff accounts can only be created through a page that is exclusively accessible by administrators.

As for customers, upon logging into the system, they will be redirected to the customer interface. The customer interface consists of a main dashboard which redirects to multiple different pages including the home, vehicle, booking, vouchers and help pages. Customers also have their own profile page which contains all of their personal information, including documents such as their identification card, matriculation card and driving licence pictures.

In the vehicles page, customers are able to view all vehicles that are available for rental and may proceed to make a booking directly. Customers can also view the details of each vehicle, including the rental amount for each individual vehicle.

The help page is a page that contains frequently asked questions to guide customers, as well as a help segment. The help segment contains the company support line and email address for any customer that would like to inquire about the system or anything related to rental bookings.

The vouchers page will display all available vouchers that can be redeemed by customers based on their booking amount. Once a voucher has been redeemed, it must be used within a specified time limit before it expires.

Other than that, customers can perform a rental booking which involves selecting a vehicle and a preferred pick-up and return location, as well as the date and time. Customers are also given the option to input a voucher code obtained from the rewards page. Following that, a full payment breakdown will be displayed and customers may proceed to pay the total rental amount via bank transfer. Once payment has been made, the customer will need to wait for staff approval in order to activate the booking. The customer may then proceed to take before-rental pictures and sign the vehicle inspection form. A few minutes before the rental period ends, customers are required to submit after-rental pictures and provide a review regarding their overall rental experience. Once the booking has been completed, the deposit will be returned to the customer within three to seven working business days.

Next, staff and administrators can log into the system, which will redirect them to their respective interface. Both staff and administrators share the same interface; however, staff are given limited access when using the system. The interface includes a main side navigation tab consisting of the Dashboard, Fleet, Bookings, Reporting, Approvals, Customer, Rewards and Hasta Staff pages. Each staff member and administrator who logs into the system will also have their own profile page which they may customize.

The dashboard page serves as an overall informational page for all staff and administrators to view. It contains key elements compiled from each page into a well-organized layout.

The fleet page allows staff to add or edit vehicle information. This includes the rental prices, vehicle documentation, vehicle service dates and other relevant details. The fleet page also includes an instant status change feature and a search function to assist staff in navigating the system more efficiently.

The bookings page displays all bookings that have been recorded in the system. Throughout the customer booking process, the system will compile and display all relevant information, including receipts, pictures, the inspection form, reviews and other details. There is also a designated field for staff to update the deposit return status once the customer's deposit has been returned.

The reporting page displays the total revenue based on the preferences set by the administrator. The data can be filtered by yearly, monthly or weekly intervals. The reporting graphs can also be adjusted by categories such as the UTM college, faculty of studies and identity within UTM. This page also offers a print feature for documentation purposes.

The approvals page is where staff will review the customer's payment receipt and activate the booking. Once a new approval request appears, a notification symbol will be displayed to notify the staff. The approval page will display all relevant information, including the booking details and the payment receipt. Once the staff member is satisfied, they may approve and activate the booking for the customer.

The customer page displays all customers that have registered within the system. Staff are able to view and verify all documents that have been uploaded by the customers. In cases where a customer is known to have caused disruption to the system, staff may change the customer's status from active to blacklisted by providing a valid reason. This will prevent the customer from creating a new booking until the blacklist issue has been resolved.

The rewards page displays all vouchers that have been used during the booking process. This page serves to monitor and document the total amount of discounts given to customers, allowing the company to understand the financial impact of vouchers. From this information, staff will also be able to determine how many vouchers should be printed in the future.

Lastly, the Hasta Staff page is exclusively accessible by administrators. This page is intended for administrators to manage the list of staff members working under them by creating accounts on their behalf. Staff information such as bank details, phone number and residential address is also stored for company purposes. Administrators are also able to create accounts for other administrators to accommodate potential future company partnerships.

1.4 Definitions, Acronyms & Abbreviations

Definitions :

Customer - One of the targeted users of the system, they are students and staff from University of Technology Malaysia (UTM).

Staff - One of the targeted users in the system, they are the staff of the owning company, Hasta Travels & Tours. Staffs have full access to the system except for viewing Reporting and Hasta Staff pages

Administrators - One of the targeted users of the system, they are the main people who represent the owning company of Hasta Travels & Tours. Administrators have full access to system without restrictions

Hasta - The short-term naming for the owning company, Hasta Travel & Tours Sdn Bhd.

Blacklist - A status assigned to a customer account when the customer is known to have caused disruption or violated the terms and conditions of the system. A blacklisted customer is prevented from creating new bookings until the issue has been resolved.

Voucher - A redeemable code owned by a customer that can be applied during the booking process to obtain a discount on the total rental amount.

Loyalty Points Points earned by customers upon completing a booking, which can be accumulated and later redeemed in exchange for vouchers or rewards.

Soft Deletion - A method of removing a record from active use by updating its status or setting a deletion timestamp, without physically removing it from the database. This preserves historical data for auditing purposes.

Cool-down Period - A mandatory one-hour gap between the end of one booking and the start of the next booking for the same vehicle, to allow for cleaning and inspection.

Acronyms :

UTM	- University Teknologi Malaysia
NRIC	- National Registration Identity Card
IC	- Identity Card
FAQ	- Frequently Asked Questions
ERD	- Entity Relationship Diagram
PK	- Primary Key
FK	- Foreign Key

2.0 Database System Requirement

2.1 System Overview

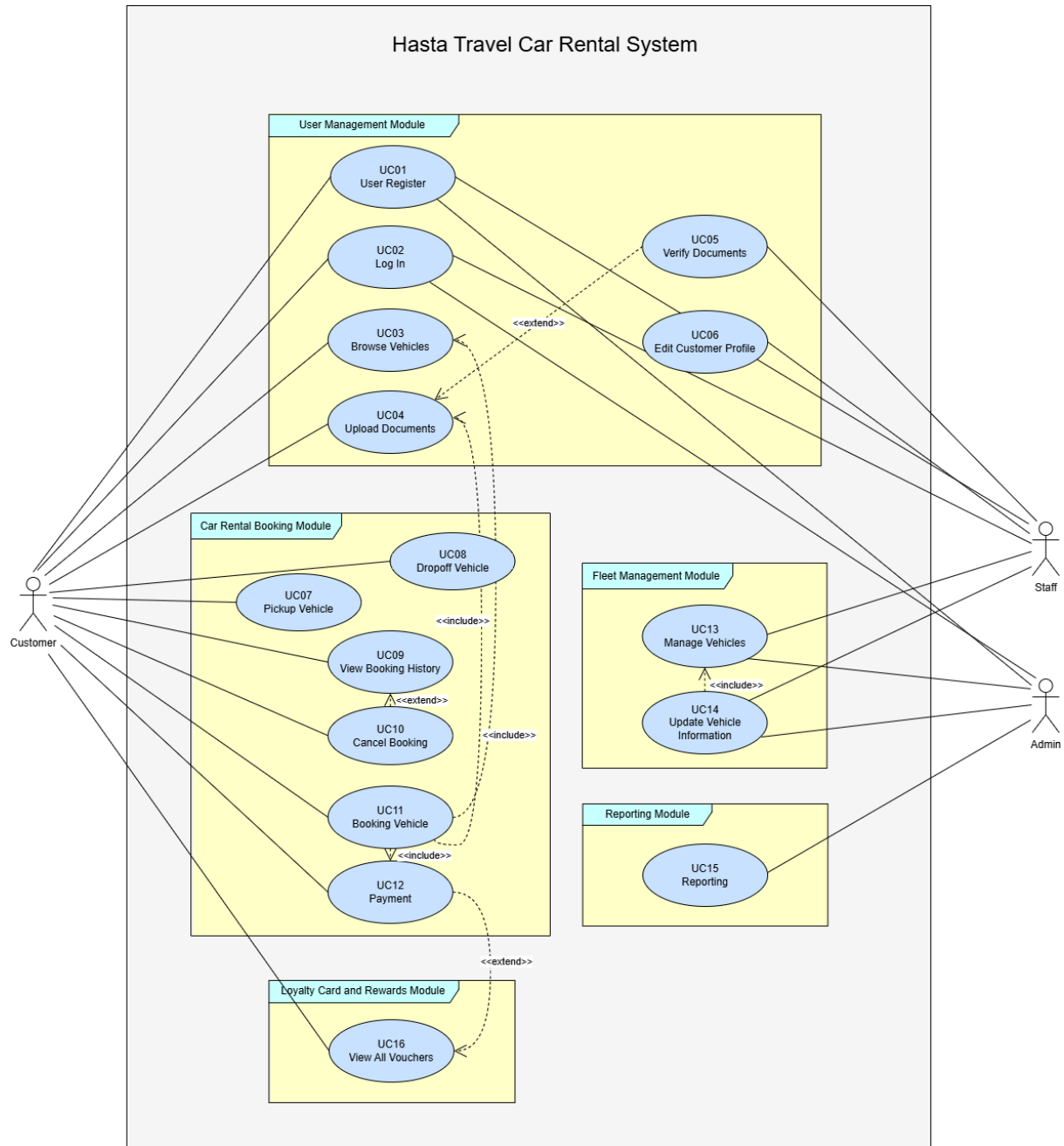


Figure : Use Case diagram

Figure shows the use case diagram for Hasta Travel System. In the diagram, three actors (small people) outside the system represent a type of user including customer, staff and admin. In the system, the use cases of users are represented by oval and group by their module. The relationship of each actor and use case is represented by a line connected between them.

Table 2.1: Description of Module and Functions for Hasta Travel Car Rental System

Use Case	Function	Modules	Description
UC01	User Register	User	This use case allows customers, staff and admin to create a new account in order to access car rental services and personalized features.
UC02	User Log In	User	This use case allows customers, staff and admin to securely log in to the system in order to access car rental services and personalized features.
UC03	Browse Fleet	User	Allows users to view and search available vehicles based on rental dates, vehicle type, and availability status.
UC04	Upload Documents	User	Allows users to upload required documents such as identification cards and driving licenses for verification before booking approval.
UC05	Verify Documents	User	Allows staff to review, approve, or reject uploaded customer documents to ensure compliance with rental requirements.
UC06	Edit Customer Profile	User	Allows staff to update customer account details.
UC07	Pickup Vehicle	Booking	Allows users to record vehicle pickup details, including time confirmation and vehicle condition status with their signs.
UC8	Dropoff Vehicle	Booking	Allows users to record vehicle pickup details, including time confirmation and vehicle condition status with their signs.
UC09	View Booking History	Booking	Allows users to view a list of their past and ongoing bookings, including rental status and payment details.

UC10	Cancel Booking	Booking	Allows users to cancel an existing booking based on system-defined cancellation policies and conditions.
UC11	Booking Vehicle	Booking	Allows users to make a car rental booking by selecting a vehicle, rental duration, and pickup/return details. This use case includes browsing vehicles, document upload, and payment.
UC12	Payment	Booking	Allows users to complete rental payments using available payment methods and submit payment confirmation for processing.
UC13	Manage Vehicles	Fleet	Allows staff or admin to manage the vehicle fleet, including monitoring availability and operational status.
UC14	Update Vehicle Information	Fleet	Allows staff or admin to update vehicle details such as price, condition, availability status, and maintenance information.
UC15	Reporting	Reporting	Allows admin to generate booking and financial reports, and view system performance analytics to support decision-making.
UC16	View All Vouchers	Rewards	Allow users to view available vouchers and loyalty cards that can be applied during booking or payment for discounts and rewards.

2.2 Reward Module

Team member responsible: Parthiv Gunalan (A24CS0178)

This module is used to manage customer loyalty program activities such as tracking points, redeeming rewards, and managing vouchers ,it is to ensure they can effectively utilize these benefits.

2.2.1 Module Description

This module manages the complete customer loyalty point for the Hasta Travel Vehicle Rental Booking System. Its purpose is to attract customers for repeat business by loyalty point reward and voucher redemption. The module allows customers to view their current loyalty point balance, view available reward, and generate unique voucher codes. It can ensure that only the valid and unexpired vouchers are applied for discounts.

Use cases included in this module:

UC14 View All Voucher and Loyalty Card:

- Customers can view their current loyalty point balance, check the redeemed history, and view a list of vouchers to check their reward status in the reward dashboard.

UC15 Redeem Reward:

- Customers can exchange accumulated loyalty points for discount vouchers such as 10% Off, 2 hours free hours. The voucher can apply on the booking page. The system validates vouchers and generates a unique voucher code upon successful booking.

UC16 Verify Voucher:

- The system will validate the voucher codes applied during the booking process. It checks the validation status, expiration dates, and usage status to prevent the invalid discount applied.

2.2.2 Transaction Requirement: Data Entry

This section supports the new reward vouchers recorded and generated into the system. Customers can choose to redeem points and require the system to generate and store accurate voucher data.

Input Data Includes:

- User identification: Customer ID that links the reward section.
- Reward selection: The type of reward chosen such as discount_10, free_hour_2.
- System-generated data:
 - Unique Voucher Code (alphanumeric string).
 - Expiry Date (automatically calculated).
 - Status (default set to 'available').
 - Redeemed At timestamp.

Constraints:

- Customer's current loyalty_points must be greater than or equal to the points_required for the selected reward.
- Generated voucher_code must be unique across the entire rewards table.
- Reward type must be selected from the predefined system configuration.
- Expiry date must be a valid future date based on system rules.

Output:

- A new voucher record is created and stored in the rewards database.
- The voucher becomes available in the customer's "My Vouchers" list.

2.2.3 Transaction Requirement: Data Update/ Deletion

This section handles the modification of loyalty points and voucher status. Physical deletion of voucher records is not supported to maintain a complete audit trail.

Used Loyalty Points :

- Trigger: When a customer successfully redeems a reward.
- Action: The system subtracting the specific points_required from the customer's current loyalty_points balance. Then, the users table updated.

Update Voucher Status after use voucher:

- Trigger: When a voucher is successfully applied to a completed booking.
- Action: The system updates the voucher record in the rewards table. The status of the voucher is changed from 'available' to 'used', and the used_at timestamp is recorded.

Output:

- The customer's point balance is immediately deducted.
- The used voucher is moved to the "Used History" tab and cannot be applied to future bookings.

2.2.4 Transaction Requirement: Data Queries

This transaction supports the retrieval and validation of loyalty data to facilitate user interaction and booking logic.

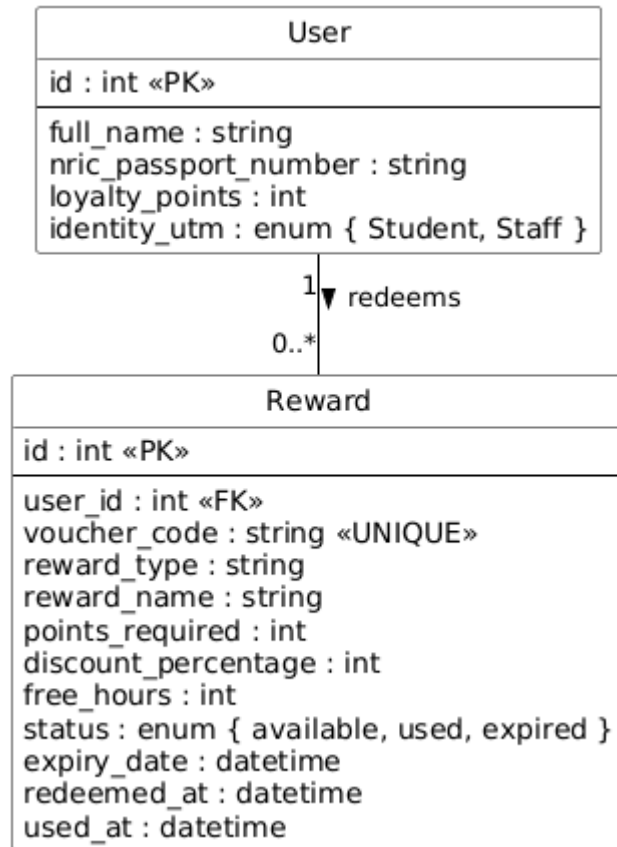
Supported Queries Include:

- Retrieve Point Balance: check the users table to display the current available points for the dashboard header.
- List Active Vouchers: Filter rewards where user_id matches the customer, status is 'available', and expiry_date is in the future.
- List History: Filter rewards where status is 'used' or expiry_date has passed.
- Validate Voucher Code: Search the rewards table by a specific code to verify its existence, ownership, validity period, and availability status during checkout.

Output:

- Real-time point balance.
- Categorized lists of vouchers
- JSON response (Valid/Invalid) with discount details for the booking engine.

2.2.5 Local Conceptual Data Model (ERD)



2.3 Fleet Management Modules

Team member responsible: HONG JIA BAO (A4CS0251)

This module is used by Staff and Admin to manage vehicles such as add, edit and view vehicle information so the Car Rental Booking module always shows correct vehicle details and availability.

2.3.1 Module Description

This module manages all vehicle information for the Hasta Travel Vehicle Rental Booking System. Its primary purpose is to maintain accurate fleet data to support reliable booking operations. The module allows staff and admin to add vehicles, update vehicle information and control vehicle availability status. It supports the vehicle lifecycle from registration to maintenance, ensuring customers are presented with valid vehicle options while minimizing booking conflict.

Use cases included in this module :

1. UC11 Manage vehicle
 - Staff or Admin can manage vehicles by adding a new vehicle, viewing vehicle list, searching and filtering vehicles, monitoring availability and operational status, deactivating vehicles when necessary to ensure only valid vehicles are offered for rental.
2. UC12 Update Vehicle Information
 - Staff and Admin can update vehicle information such as rental price packages, mileage, condition notes, availability status, and maintenance, insurance and road tax details. The use case also supports updating related documents and images to keep fleet records complete and accurate.

2.3.2 Transaction Requirement: Data Entry

This section supports the registration of new vehicles into the system. Staff or admin are required to enter complete vehicle information to ensure accurate fleet records.

Input Data Includes :

- Vehicle details such as car brand, car model, year, color , plate number
- Vehicle specifications such as car type, transmission type, fuel type, number of seats
- Rental information such as rental rate per hour, mileage, pricing packages.
- Operational status such as available, in use, unavailable, maintenance.
- Supporting information such as description
- Uploaded files such as vehicle image, road tax document, insurance document
- Maintenance information : last service date, next service date, document expiry date

Constraints :

- Plate number must be unique for each vehicle.
- Rental rates, mileage, pricing values must be valid numeric values.
- Status must be selected from predefined system values.
- Uploaded files must follow allowed formats and size limits
- Service and expiry dates must follow logical date constraints.

Output :

- A new vehicle record is created and stored in the fleet database.
- The vehicle becomes available for viewing and booking based on its availability status.

2.3.3 Transaction Requirement: Data Update/ Deletion

This section allows staff and admin to update existing vehicle information to ensure fleet data remains accurate and up to date throughout the vehicle lifecycle. Physical deletion of vehicle records is not supported in this module.

Update Vehicle Information :

Trigger when a staff and admin edits vehicle details. The system updates the vehicle record in the cars table with new information, including :

- Vehicle identification details and specifications
- Rental pricing and pricing packages
- Mileage and condition description
- Maintenance schedules and document expiry information
- Vehicle images and compliance documents

Update Vehicle Status :

Trigger when a vehicle is no longer available for rental. Instead of deleting the vehicle record, the system performs a soft delete by updating the vehicle's status attribute to unavailable. This ensures booking history remains intact and requirements can still be fulfilled.

Output :

The vehicle is excluded from active rental selection but remains stored in the database for reference and historical purpose.

2.3.4 Transaction Requirement: Data Queries

This transaction supports retrieval and analysis of fleet data to assist daily operations and decision-making.

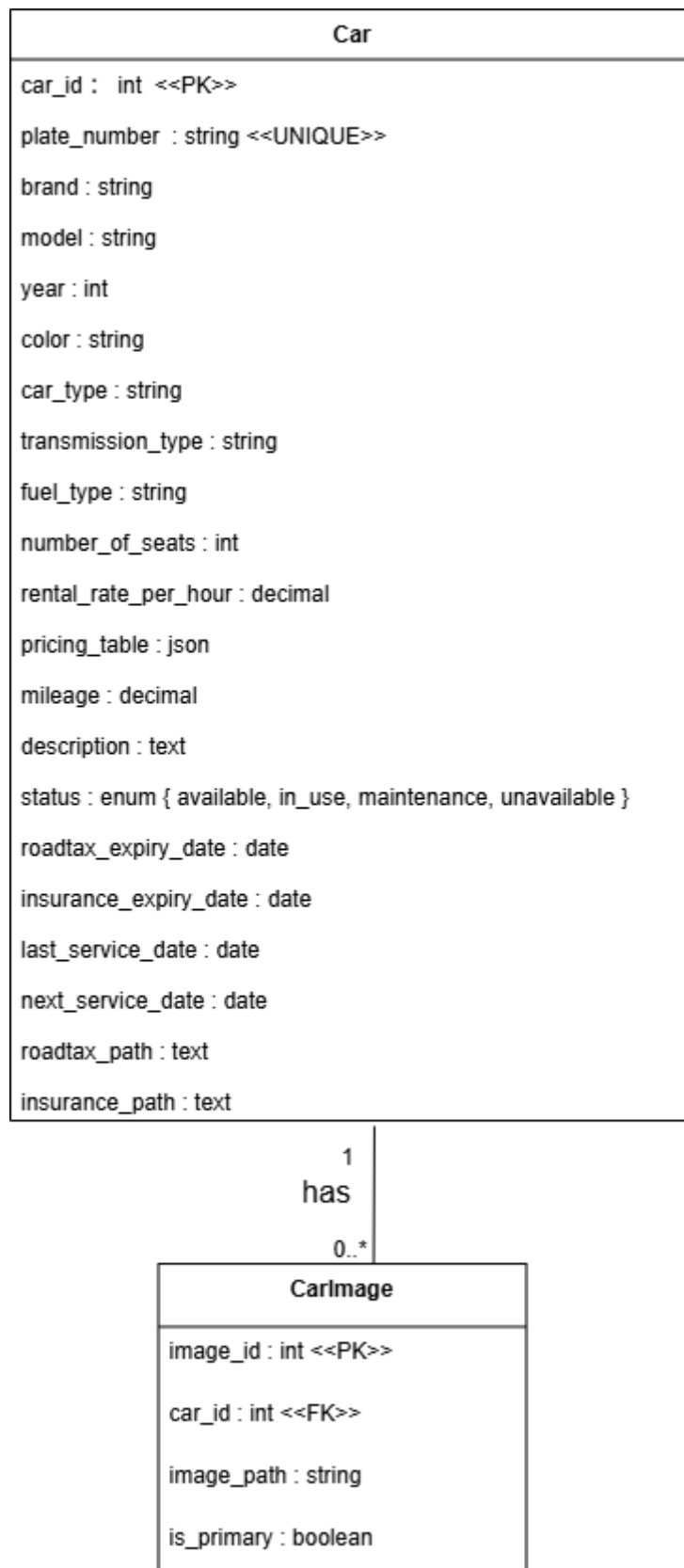
Supported Queries Include :

- View the complete list of vehicles in the system.
- Search vehicles by brand, model, plate number, or vehicle type.
- Filter vehicles by availability status such as available, in use, maintenance, unavailable.
- View detailed information of a selected vehicle, including pricing, documents, images.
- Retrieve fleet statistics such as total vehicles and vehicle counts by status.

Output :

- Filtered vehicle listings.
- Detailed vehicle information for operational view
- Summary statistics to support fleet monitoring and management decisions

2.3.5 Local Conceptual Data Model (ERD)



2.4 User Management Module

Team member responsible: LEONG JIA LING (A24CS0104)

The User Management Module manages user authentication, profile information and identify verification in the Hasta Travel Car Rental System. It ensures that only verified users can access system functions based on their roles, supporting reliable operation of other system modules.

2.4.1 Module Description

This module manages user accounts and identifies verification for the Hasta Travel Car Rental System. Its primary purpose is to ensure that only valid users can access the system functions, while maintaining accurate personal data and supporting document-based verification important to booking approval. The module supports user lifecycle activities including registration and login, profile management, document upload, staff verification and admin verification. The module also includes password and account security management. Users are required to verify their email during registration. Passwords are stored securely using encryption or hashing methods. The system supports password reset and forgot-password functions to help users recover their accounts securely. To prevent unauthorized access, the system may temporarily lock accounts after multiple failed login attempts and enforce session timeouts for inactive users. The module applies role-based access control to ensure users can only perform actions permitted for their roles. Customers are allowed to manage their own profiles and upload documents. Staff are allowed to review documents and update verification status. Admins have broader privileges such as managing user roles, viewing all user records and controlling blacklist status. This access control ensures system security and proper governance. It also supports user status control such as active, pending and blacklisted accounts to reduce risk and ensure that only eligible users can book vehicles.

Use cases included in this module :

1. UC01 User Register and Log in
 - Customers , staff and admin log in to the system and access features based on their assigned role.
2. UC03 Upload Documents
 - Customers upload identity documents such as IC/Passport, matric/staff card, and driving license for verification.
3. UC04 Verify Documents
 - Staff or admin verify uploaded documents and update verification status to ensure compliance with rental requirements.
4. UC05 Edit Customer Profile
 - Customers update profile information while staff or admin may update customer details for data accuracy and account administration.

2.4.2 Transaction Requirement: Data Entry

This section supports the creation of new user accounts and initial profile data entry.

Customers are required to submit personal and contact information to allow the system to build an authorized identity record.

Input Data Includes :

- Personal details such as full name, NRIC or passport number, nationality, UTM identity (Student or Staff), matric or staff card number.
- Contact details such as email, country code for phone number, contact number, password.
- Address details such as mailing address, hostel info, college name, faculty.
- Roles such as staff or admin
- Status such as active or pending or blacklisted
- Supporting documents: IC or Passport, matric or staff card and driving license

Constraints :

- Email must be unique.
- NRIC or passport number must be unique.
- Matric or staff card number must be unique
- Password must follow minimum length requirement and stored securely
- Roles must follow predefined values such as customer or admin and cannot be arbitrary.
- Uploaded files must follow allowed format and size

2.4.3 Transaction Requirement: Data Update/ Deletion

This section supports updating user-related status and verification information to control account eligibility. Staff and admin are not allowed to modify customer personal profile data. Instead, they are only allowed to verify customer documents and update customer account status. Physical deletion of user records is not required for normal operation. Therefore, the module uses soft deletion and status management.

Update User Profile

Trigger when a customer updates their own profile. The system allows customers to update the following information :

- personal, contact, and address information
- driving license details such as license number and expiry date
- bank details for refund
- emergency contact details
- uploaded document paths such as IC or Passport, matric card, driving license

Update Verification Status

Trigger when staff or admin performs document verification. The system updates `is_verified` to true or false.

Update Customer Status

Trigger when staff or admin changes customer account status.

- If status becomes blacklisted, the system stores `blacklist_reason`, sets `is_active` to false, and cancels all pending, approved, or active bookings to prevent operational risk.
- If status becomes active, the blacklist reason is cleared and customer access to the system is restored.

Soft Deletion

Trigger when a user record should not be physically removed from the system. The system uses `deleted_at` for soft deletion. The record remains stored in the database for historical reference and audit purposes.

Audit Trail and Logging

The system maintains an audit trail for accountability and monitoring purposes. Actions such as document verification, status updates, blacklisting, and profile changes are logged with timestamps and the responsible staff or admin ID. These records support dispute resolution, monitoring and system transparency.

Output :

- Updated customer verification and account status are stored in the system.
- Verification and blacklist actions immediately affect booking eligibility and system access.
- Soft-deleted users are excluded from normal active user listings but remain stored for reference.

2.4.4 Transaction Requirement: Data Queries

This section supports retrieving user information for daily operations, verification workflow, and user account administration.

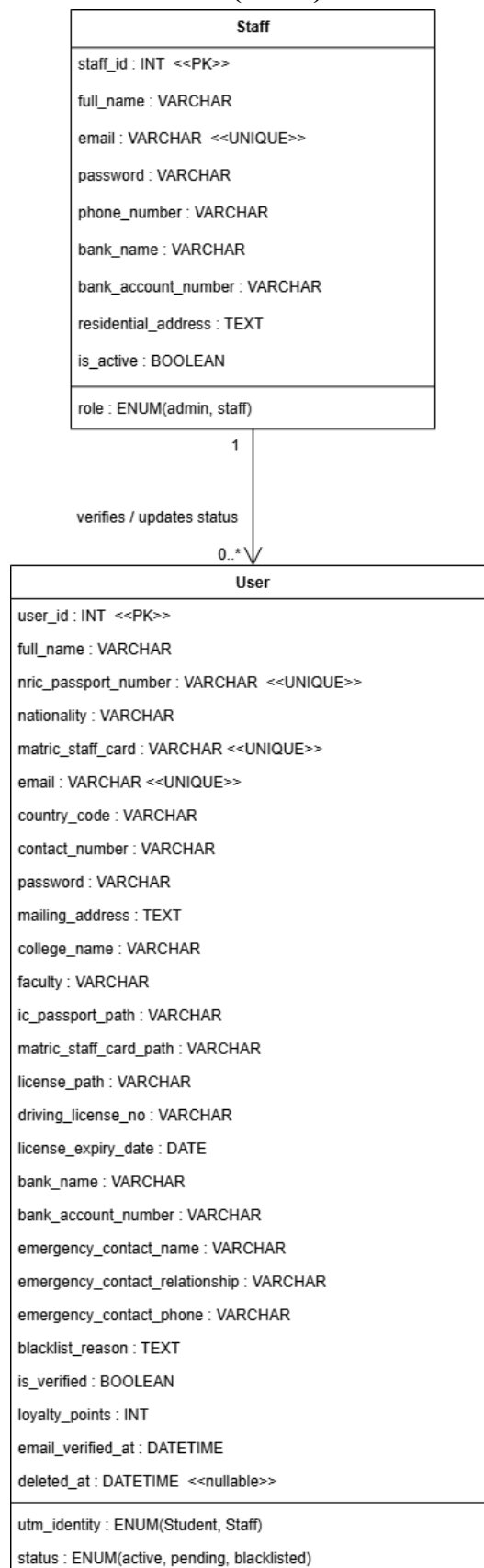
Supported Queries Include :

- View list of customers with pagination
- Search customers by name, email, NRIC or passport, matric or staff card, contact number.
- Filter customers by status such as active or pending or blacklisted) , identity type such as Student or Staff, verification state such as verified or unverified
- View a specific customer profile in detail including booking history summary
- Retrieve dashboard statistics such as total customers, verified count, blacklisted count.

Output :

- Filtered customer lists for admin and staff operations.
- Detailed customer profile data for verification and management.
- Summary statistics for monitoring and decision-making.

2.4.5 Local Conceptual Data Model (ERD)



Admin is represented as a role within the Staff entity (role=admin), not a separate table.

2.5 Booking Module

Team member responsible: NOAH HENG ZHI QIANG (A24CS0081)

This module is used by Customers to book vehicles and process transactions, it is used to manage all the booking processes such as vehicle availability and all rental phases.

2.5.1 Module Description

This module manages the complete rental lifecycle for the Hasta Travel Vehicle Rental Booking System. Its purpose is to provide booking reservations and show the whole rental handover process clearly. The module allows customers to check real-time vehicle availability, calculate dynamic rental costs, and submit proof of payment. It supports the entire booking process, it makes sure the system follows business rules such as minimum rental durations and maintenance cool-down periods.

Use cases included in this module:

1. UC02 Browse Fleet
 - Customers browse vehicles as a pre-booking activity.
2. UC09 Booking Vehicle
 - Customers can search vehicles by date, view calculated costs, and they can submit a booking request. The system performs real-time validation to prevent overlapping schedules.
3. UC10 Payment
 - Once customers submit their reservation, they need to submit the payment proofs. The system stores this proof for staff verification.
4. UC06 Pickup & Return Vehicle
 - Customers need to upload vehicle condition photos and sign rental agreements before pickup. After that ,they need to submit return evidence upon completion.
5. UC08 Cancel Booking
 - Customers can cancel an existing reservation 24 hours before start time, system updates the booking status and timestamp recording.

2.5.2 Transaction Requirement: Data Entry

This section supports the creation of new rental reservations into the system. Customers are required to enter trip details and upload financial proof to initiate a booking.

Input Data Includes:

- Trip details: Start date, end date, pickup location, dropoff location.
- Vehicle selection: Selected car_id derived from the user's choice.
- Financial information: System-calculated rental_amount, delivery_fee, voucher_discount, and grand_total.
- Supporting documents: Uploaded payment receipt image file.
- User identification: Customer id.

Constraints:

- Availability: Users cannot select the date for the car that is currently at status "active", "approved", or "pending".
- Cool-down Period: New booking only can start 1-hour after the previous booking's end time.
- Date Logic: Return time must be after the pickup time.
- File Validation: Uploaded receipt must be an image format such as JPG, PNG and not exceed 2MB.
- Location: Pickup/Dropoff locations must match valid service areas for delivery fee calculation.

Output:

- A new booking record with status set to 'Pending' is created in the bookings database.
- The selected vehicle slot is reserved to prevent duplicate booking attempts.

2.5.3 Transaction Requirement: Data Update/ Deletion

This section allows the system and customers to modify the booking state. Physical deletion of booking records is restricted to maintain audit history. All the booking cancellations are handled through status updates.

Update Booking Status when Cancellation:

- Trigger: When a customer cancels the booking.
- Action: The system updates the booking record's status to 'cancelled' and records the cancelled_at timestamp. After that, the car releases for other customers.

Update Booking Details when Pick Up and Drop off of Car:

- Trigger: When a customer uploads documents for pick up or dropoff process
- Action: The system updates the booking record with:
 - Pick up car: JSON paths for pickup_photos, agreement_signed_path, and agreement_signed_at timestamp.
 - Drop off car : JSON paths for return_photos, review_rating, and review_comment.

Update Loyalty point:

- Trigger: When a booking status changes to 'completed'.
- Action: The system updates the users table to increment the loyalty_points balance.

Output:

- The booking status reflects the current stage (e.g., Active, Completed, Cancelled).
- Vehicle availability is dynamically adjusted based on the updated status.

2.5.4 Transaction Requirement: Data Queries

This transaction supports the retrieval of booking data to enforce business logic and provide user dashboards.

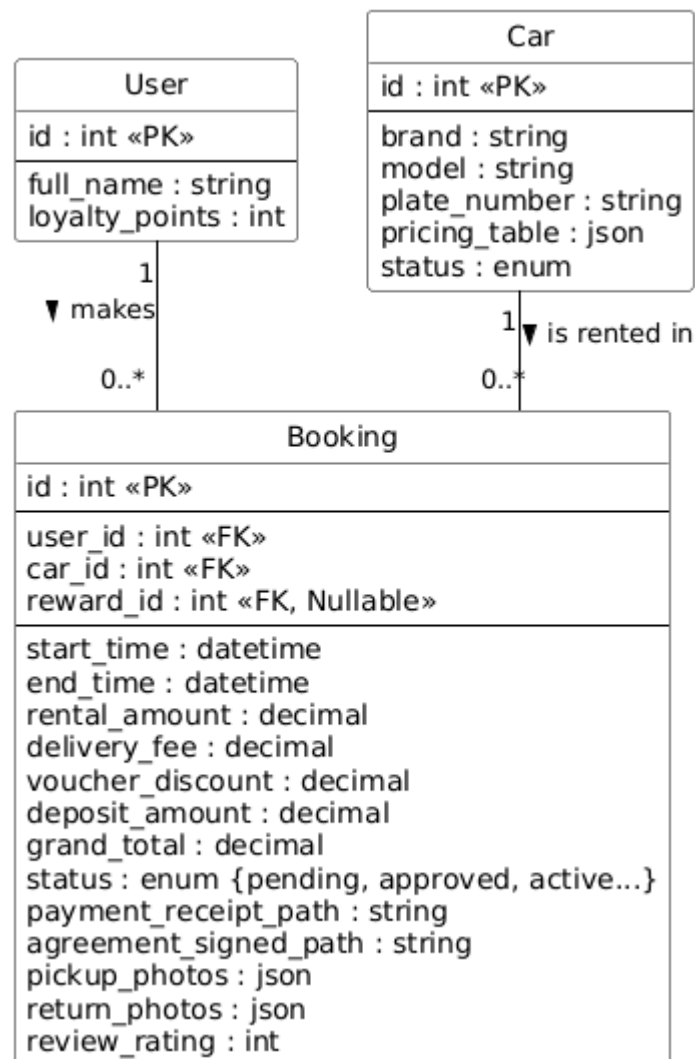
Supported Queries Include:

- Check Availability : Query the bookings table to find overlaps between requested dates and existing bookings, and add a 1-hour cool-down buffer to the previous booking's end time.
- Calculate Dynamic Pricing: Retrieve the pricing_table (JSON) from the cars table to determine the price if the requested duration under the package rate or standard hourly rate.
- View Booking History: Retrieve all bookings of users, then categorise all the booking into "Current" and "History" lists.
- Retrieve Booking Details: Obtain all the booking details by booking ID, including linked car information, uploaded photo paths, and payment status.

Output:

- Boolean status includes Available or Unavailable for date selection.
- Calculated price breakdown for the frontend display.
- Lists of bookings for the customer dashboard.

2.5.5 Local Conceptual Data Model (ERD)



2.6 Reporting & Analytics Module

Team member responsible: LEE PEI YUAN A24CS0262

This module is used by Admin to visualize financial performance , track monthly total sales and the booking information, this feature helping them plan better promotions and manage their cars more effectively.

2.6.1 Module Description

This module provides an administrative view into the performance of the Hasta Travel Vehicle Rental Booking System. Its purpose is to combine and analyse the operational data into meaningful financial and statistical reports. Admins can track total sales revenue over specific timeframes such as daily, monthly and weekly, it analyses customer profiles by categories such as Faculty, College, and Identity. This makes admin easier planning the business strategy by finding peak rental times and important customer segments without needing any manual paperwork.

Use cases included in this module

1. UC13 Reporting
 - Admin can access a dashboard to generate dynamic financial and statistical reports. This includes "Income Reports" that is used to view sales trends and "Statistical Reports" that is used to view main user distribution. The system displays charts, graphs, and summary tables in real time.

2.6.2 Transaction Requirement: Data Entry

This section supports the creation of analytical views for business. This module does not create any new persistent records but it requires specific input parameters that are obtained from other modules to generate dynamic datasets for visualization.

Input Data Includes:

- Report Type Selection: Choice between 'Daily', 'Monthly', or 'Booking Info'.
- Timeframe Parameters: Specific year for monthly reports or a specific date reference for weekly breakdowns.
- Category Filters: Filters for statistical analysis, such as 'Faculty and College', 'Student', or 'Staff'.

Constraints:

- Year Validation: The selected year must contain existing booking records.
- Date Format: Date must follow valid calendar formats (YYYY-MM-DD).
- Access Control: Only the 'Admin' role can generate these reports.

Output:

- A dynamically generated visual report that contains charts or tables is display on the dashboard.
- Aggregated statistical data such as total sales and total amount of booking is calculated and displayed.

2.6.3 Transaction Requirement: Data Update/ Deletion

This module functions as a Read-Only analytical tool for decision support. It does not support the direct modification of booking or user records, it ensures data integrity for auditing purposes.

Data Integrity:

- The system ensures that the reports reflect the current state of the database. The modules automatically reflected any updates or deletions that were made in the booking or user. This module does not provide interfaces for altering the source data.

2.6.4 Transaction Requirement: Data Queries

This transaction supports the complex aggregation and calculation of large datasets to derive meaningful information for business planning.

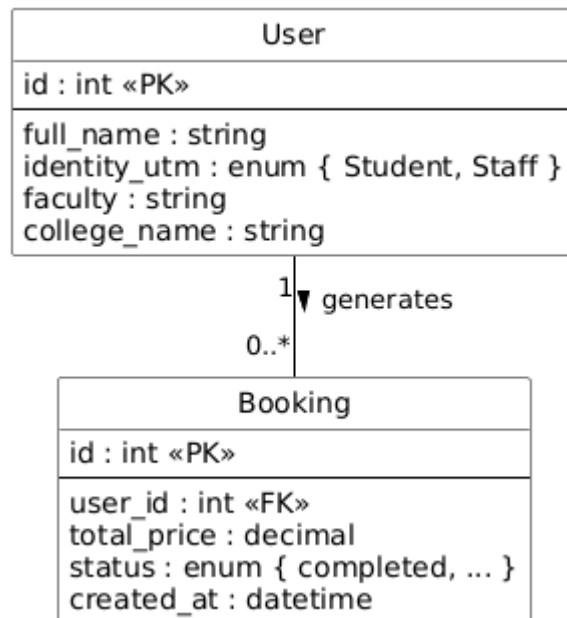
Supported Queries Include:

- Generate Income Report (Daily/Monthly): The system pulls data from the bookings table to calculate total earnings and the number of sales. It organizes these results by DATE(created_at) for weekly views or MONTH(created_at) for annual views and only include bookings that are already marked as 'completed'
- Analyze booking population(Faculty/College): The system pulls the data from bookings and users tables. The system groups the results by users.faculty or users.college_name to arrange the highest student segments.
- Analyze Identity Split: Query the dataset to compare booking volumes between 'Student' and 'Staff' identities.
- Calculate Global Totals: Provide a financial summary by retrieving the sum of all total_sales and total_amount for the displayed report.

Output:

- Chart Datasets: The frontend uses charts, such as bar charts for revenue and pie charts for customer types, to visualize the analysis.
- Tabular Lists: Arranged lists of Faculties and Colleges by booking volume.
- Financial Summaries: Total revenue chart for the selected period.

2.6.5 Local Conceptual Data Model (ERD)



3.0 Global Conceptual Data Model (Global ERD)

The Global Conceptual Data Model (Global ERD) refines and integrates the local conceptual data models defined in the previous chapter, Database System Requirement, covering the User Management, Fleet Management, Car Rental Booking, Rewards and Reporting modules. The Global ERD consists of 6 entity types and 8 relationships, providing a detailed snapshot of how the data in Hasta Travel Vehicle Rental Booking System is stored and interrelated in the system. By combining the individual module requirements, this model resolves the potential conflicts and redundancies serving as the blueprint for physical database implementation. It ensures that all critical data flows ranging from user verification and fleet management to complex booking transactions and loyalty rewards are effectively supported while maintaining the strict data integrity.

3.1 Glocal Conceptual Design ERD (Lucidchart)

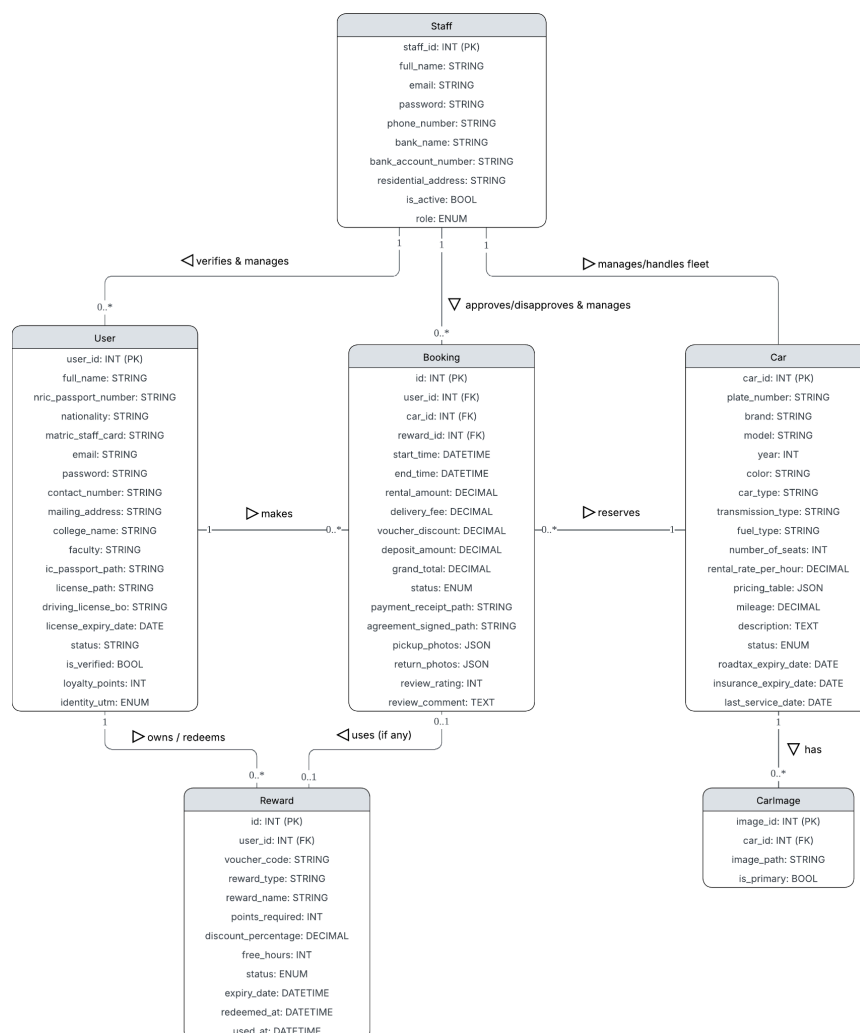


Figure 3.1 : Global Conceptual Entity Relationship Diagram

3.2 Verified Glocal Conceptual Design (Verified ERD) (Lucidchart)

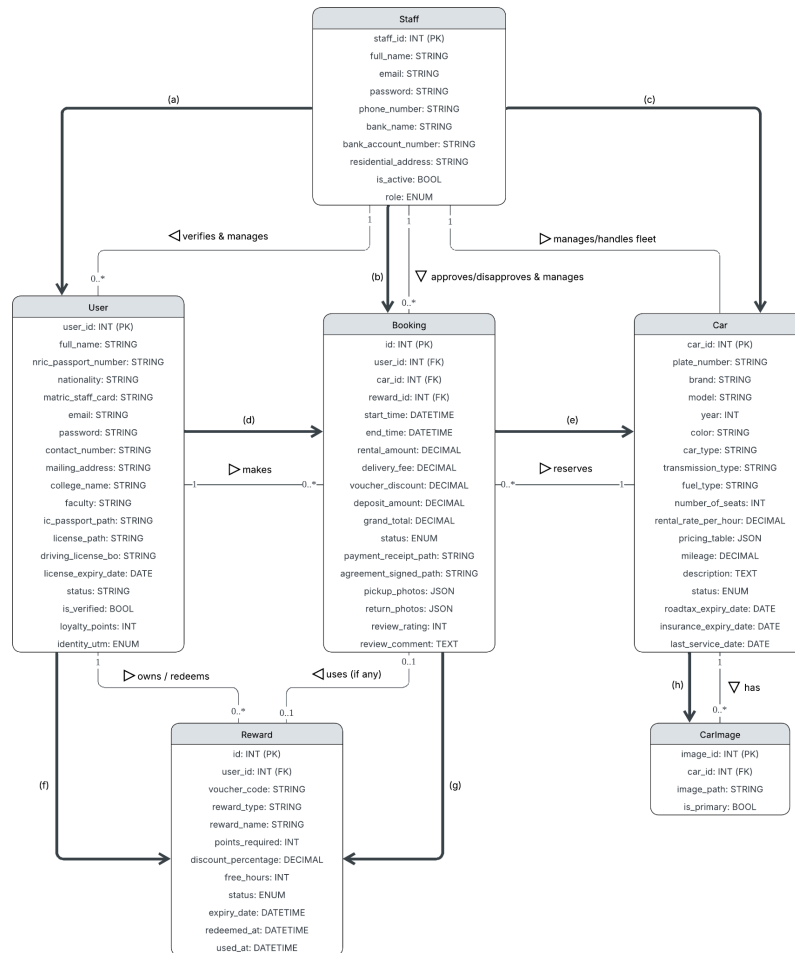


Figure 3.2 : Verified Global Conceptual Entity Relationship Diagram

Symbol	Transaction	Description
(a)	2.4.3 (Data Update), 2.4.4	Staff verifies the customer documents and if the documents are valid, he/she updates the verification status (is_verified). Staff manages the user account and updates the account status from active to blacklisted if a user has violated the terms and conditions. Staff also views and searches user lists to see the details of customers.
(b)	2.6.4	Admin accesses the booking records to generate the income reports and analyze the booking trends.
(c)	2.3.2, 2.3.3 (Data Update), 2.3.4	Staff enters new vehicle details, updates fleet information like pricing and availability status and checks the fleet list to adjust the records as needed.
(d)	2.5.2, 2.5.3 (Data Update), 2.5.4	The user (customer) creates a new booking, cancels an existing booking if there is 24 hours before the start time of the booking and view their personal booking history.
(e)	2.5.2, 2.5.4	Booking validates the vehicle availability, reserves a specific car slot to prevent overlapping booking and retrieves the pricing details of the vehicle for calculation of total amount for booking.
(f)	2.2.2, 2.2.3 (Data Update), 2.2.4	The user (customer) redeems loyalty points for rewards and the system deducts the points from the user's loyalty points balance and displays all available vouchers.
(g)	2.2.3 (Data Update), 2.2.4, 2.5.2	Valid voucher code is applied to a booking, the discount is validated and the voucher status is updated to 'Used' upon booking completion.
(h)	2.3.2, 2.3.3 (Data Update)	During vehicle registration or update, multiple images files are uploaded and linked to a specific vehicle record for display purposes.

3.3 Data Dictionary for the global conceptual ERD

3.3.1 Description of Entities

Entity Name	Description	Aliases	Occurrence
User	Represents the customers either UTM students or UTM staff who register and undergo the verification process in order to book a vehicle rental.	Customer	Each occurrence represents a registered customer account containing the personal data either being verified, not yet verified or blacklisted. The entity can serve as a data source for the Reporting Module by allowing the system to group the user details like faculty, college and identity type to create reports and insights about the customer base.
Staff	Represents the administrative and operational personnel who are responsible for managing the fleet, verifying the users and approving as well as handling the bookings.	Admin, Employee/Staff	Each occurrence represents a staff with specific access control or rights. The Admin defined within this entity are authorized to access the Reporting Module dashboard to view financial reports, system analytics and alter the details of a staff.
Car	Represents the specifications, status or availability of the vehicles in the system.	Vehicle	Each occurrence represents a vehicle in the fleet inventory. The records of fleet allow the system to track the usage of a vehicle and can be used to help the admins analyze the performance and vehicle availability trends.
Booking	Represents rental records and transactions which links a user to a specific vehicle/car for a specific duration of time.	Reservation	Each occurrence represents a booking record from the initial request to completion. The entity provides the Reporting Module with key details like the total payment and timestamps that are used to calculate the monthly sales and generate the

			income reports.
Reward	Represents loyalty vouchers or coupons that are owned by users and can be redeemed using loyalty points.	Voucher	Each occurrence represents a specific redeemable voucher owned by a user, The records track the discount applied to bookings, ensuring that the value of each booking reflects the applied discount and accurately shows the user's savings.
CarImage	Represents the images associated with a specific vehicle for display purposes.	Vehicle Photo	Each occurrence represents an image file path linked to a specific car, allowing the system to display multiple angles of a vehicle in the fleet without cluttering the main Car table

3.3.2 Description of Relationships

Entity 1	Multiplicity	Relationship	Multiplicity	Entity 2	Description
User	1	makes	0..*	Booking	A single user can make multiple bookings.
Staff	1	verifies and manages	0..*	User	A staff can verify documents of multiple users and manage multiple user accounts.
Staff	1	approves or disapproves and manages	0..*	Booking	A staff can process multiple bookings either approving or rejecting booking requests sent by the users and manage the booking process.
Staff	1	manages or handles fleet	0..*	Car	A staff can add, update and change the availability status of multiple vehicles in the fleet inventory.
Booking	0..*	reserves	1	Car	Each booking reserves exactly a specific car from the fleet ensuring that the vehicle is booked for the selected time duration.
User	1	owns or redeems	0..*	Reward	A user can redeem and own multiple loyalty vouchers or rewards.
Booking	0..*	Uses (if any)	0..1	Reward	A booking may apply a voucher or reward for discount
Car	1	has	0..*	CarImage	A vehicle or car can be associated with multiple images.

3.3.2 Description of Attributes

Entity Name	Attributes	Data Type	Constraint	Description
User	user_id	INT	Primary Key (PK), Auto Increment	Unique identifier for the user
	full_name	STRING	Not Null	Full name of the user
	nrhc_passport_number	STRING	Unique, Not Null	Unique NRIC or Passport number
	nationality	STRING	Not Null	Nationality of user
	matric_staff_card	STRING	Unique, Not Null	Unique Matric ID of student or Staff ID of staff
	email	STRING	Unique, Not Null	Registered email address (login credential)
	password	STRING	Unique, Not Null	Hashed login password (hashed for security)
	contact_number	STRING	Not Null	User mobile phone number
	mailing_address	STRING	Not Null	Permanent residential or mailing address
	college_name	STRING	Not Null	Name of residential college inside or outside UTM (if applicable)
	faculty	STRING	Nullable	Name of the faculty (if applicable)
	ic_passport_path	STRING	Nullable	File path to the uploaded NRIC/Passport document image
	license_path	STRING	Nullable	File path to the uploaded driving license image
	driving_license_no	STRING	Nullable	Unique driving license

				number
	license_expiry_date	DATE	Nullable	Expiry date of the driving license
	status	STRING	Enum: 'Active', 'Pending', 'Blacklisted' (Default: 'Active')	Account status (e.g. Active, Blacklisted)
	is_verified	BOOL	Default: False	Flag indicating if the documents of user have been verified
	loyalty_points	INT	Default: 0	Current loyalty points earned by user
	identity_utm	ENUM	Enum: 'Student', 'Staff'	User type: "Student" or "Staff"
Staff	staff_id	INT	Primary Key (PK), Auto Increment	Unique identifier for the staff
	full_name	STRING	Not Null	Full name of staff
	email	STRING	Unique, Not Null	Staff login email address
	password	STRING	Unique, Not Null	Hashed login password (hashed for security)
	phone_number	STRING	Not Null	Staff mobile phone number
	bank_name	STRING	Not Null	Staff's bank name
	bank_account_number	STRING	Not Null	Staff's bank account number
	residential_address	STRING	Not Null	Staff's home or residential address
	is_active	BOOL	Default: True	Flag to grant or revoke the access control to the system
	role	ENUM	Enum:	Access level: "Admin"

			'Admin', 'Staff' (Default: 'Staff')	or "Staff"
Car	car_id	INT	Primary key (PK), Auto Increment	Unique identifier for the vehicle
	plate_number	STRING	Unique, Not Null	Unique vehicle registration number
	brand	STRING	Not Null	Vehicle manufacturer
	model	STRING	Not Null	Vehicle model name
	year	INT	Not Null	Manufacturing year of vehicle
	color	STRING	Not Null	Vehicle color
	car_type	STRING	Not Null	Category of vehicle (e.g. Sedan, Hatchback)
	transmission_type	STRING	Not Null	Gear type (e.g. Auto, Manual)
	fuel_type	STRING	Not Null	Fuel type (e.g. Petrol, Diesel)
	number_of_seats	INT	Not Null	Seating capacity of the vehicle
	rental_rate_per_hour	DECIMAL	Not Null	Hourly rental rate/cost
	pricing_table	JSON	JSON, Nullable	Complex pricing rules
	mileage	DECIMAL	Not Null	Current mileage or odometer reading
	description	TEXT	Nullable	Description of vehicle
	status	ENUM	Enum: 'Available', 'In Use', 'Maintenance', 'Unavailable' (Default: 'Available')	Availability status of vehicle: "Available", "In Use" "Maintenance" or "Unavailable"

	roadtax_expiry_date	DATE	Nullable	Expiry date of vehicle roadtax
	insurance_expiry_date	DATE	Nullable	Expiry date of vehicle insurance
	last_service_date	DATE	Nullable	Most recent service/maintenance date
CarImage	image_id	INT	Primary Key (PK), Auto Increment	Unique identifier for the image
	car_id	INT	Foreign Key (FK) referencing Car (car_id), On Delete Cascade	Reference to the associated Car
	image_path	STRING	Not Null	File path where the image is stored
	is_primary	BOOL	Default: False	Flag indicating if the image is the main image used for display purpose
Reward	id	INT	Primary Key (PK), Auto Increment	Unique identifier for the reward
	user_id	INT	Foreign Key (FK) referencing User (user_id), On Delete Cascade	Reference to the User who owns the reward
	voucher_code	STRING	Unique, Not Null	Unique code used for voucher redemption to get the discount
	reward_type	STRING	Not Null	Type of the reward (e.g. 'discount_10' stands for 10% discount)
	reward_name	STRING	Not Null	Display name of the reward/voucher

	points_required	INT	Not Null	Cost in loyalty points to redeem the reward/voucher
	discount_percentage	DECIMAL	Nullable	Percentage of discount for voucher
	free_hours	INT	Nullable	Number of free hours granted (if applicable)
	status	ENUM	Enum: 'Available', 'Used', 'Expired' (Default: 'Available')	Validity status of voucher: "Available", "Used" or "Expired"
	expiry_date	DATETIME	Not Null	Date when the voucher becomes invalid
	redeemed_at	DATETIME	Not Null	Timestamp when loyalty points are exchanged to redeem new voucher/reward
	used_at	DATETIME	Nullable	Timestamp when voucher is applied to booking
Booking	id	INT	Primary Key (PK), Auto Increment	Unique identifier for the booking record
	user_id	INT	Foreign Key (FK) referencing User (user_id)	Reference to User who made the booking
	car_id	INT	Foreign Key (FK) referencing Car (car_id)	Reference to Car being rented
	reward_id	INT	Foreign Key (FK) referencing Reward (id), Nullable	Reference to Reward used (if any)
	start_time	DATETIME	Not Null	Scheduled vehicle pickup date and time

	end_time	DATETIME	Not Null	Schedule vehicle dropoff date and time
	rental_amount	DECIMAL	Not Null	Rental amount for selected vehicle and time
	delivery_fee	DECIMAL	Default: 0.00	Additional fee for pickup or return delivery for location other than Student Mall
	voucher_discount	DECIMAL	Default: 0.00	Amount deducted by the applied voucher (if any)
	deposit_amount	DECIMAL	Not Null	Deposit that will be refunded after completing a booking if there is no issue or damage on vehicle rented
	grand_total	DECIMAL	Not Null	Final total amount to be paid
	status	ENUM	Enum: 'Pending', 'Approved', 'Active', 'Completed', 'Cancelled', 'Rejected' (Default: 'Pending')	State of booking: "Pending", "Approved", "Active", "Completed", "Cancelled" or "Rejected"
	payment_receipt_path	STRING	Nullable	File path to the payment receipt image
	agreement_signed_path	STRING	Nullable	File path to the signed rental agreement
	pickup_photos	JSON	JSON, Nullable	Array of image paths taken at pickup location
	return_photos	JSON	JSON, Nullable	Array of image paths taken at return location
	review_rating	INT	Nullable,	Customer rating in

			Integer (1-5)	stars
	review_comment	TEXT	Nullable	Customer feedback text

4.0 Appendices

4.1 Use Case Diagram for the system

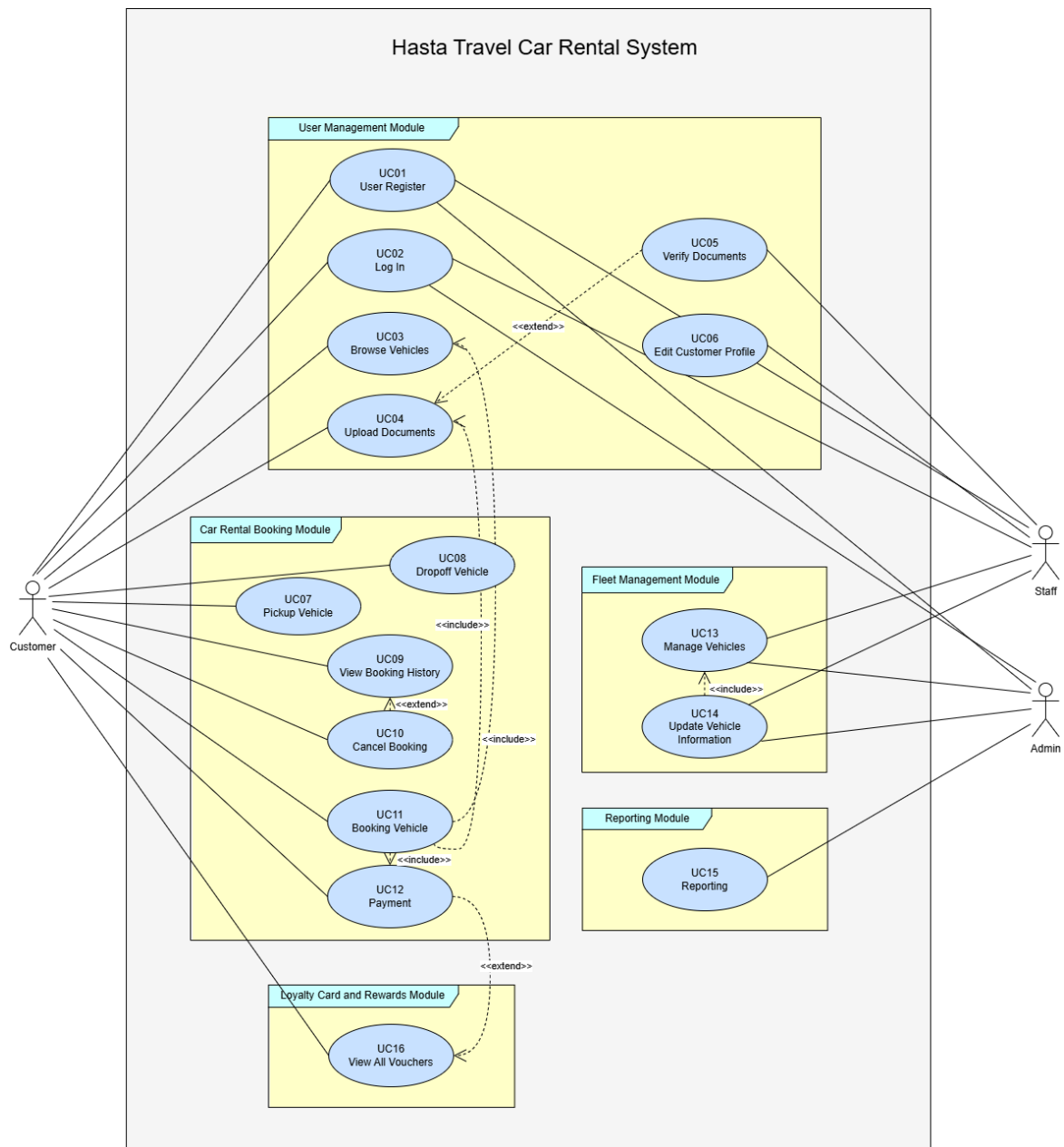


Figure 4.1: Use Case Diagram for Hasta Vehicle Rental Booking System

4.2 Activity Diagrams for every user case identified

4.2.1 UC01: User Register

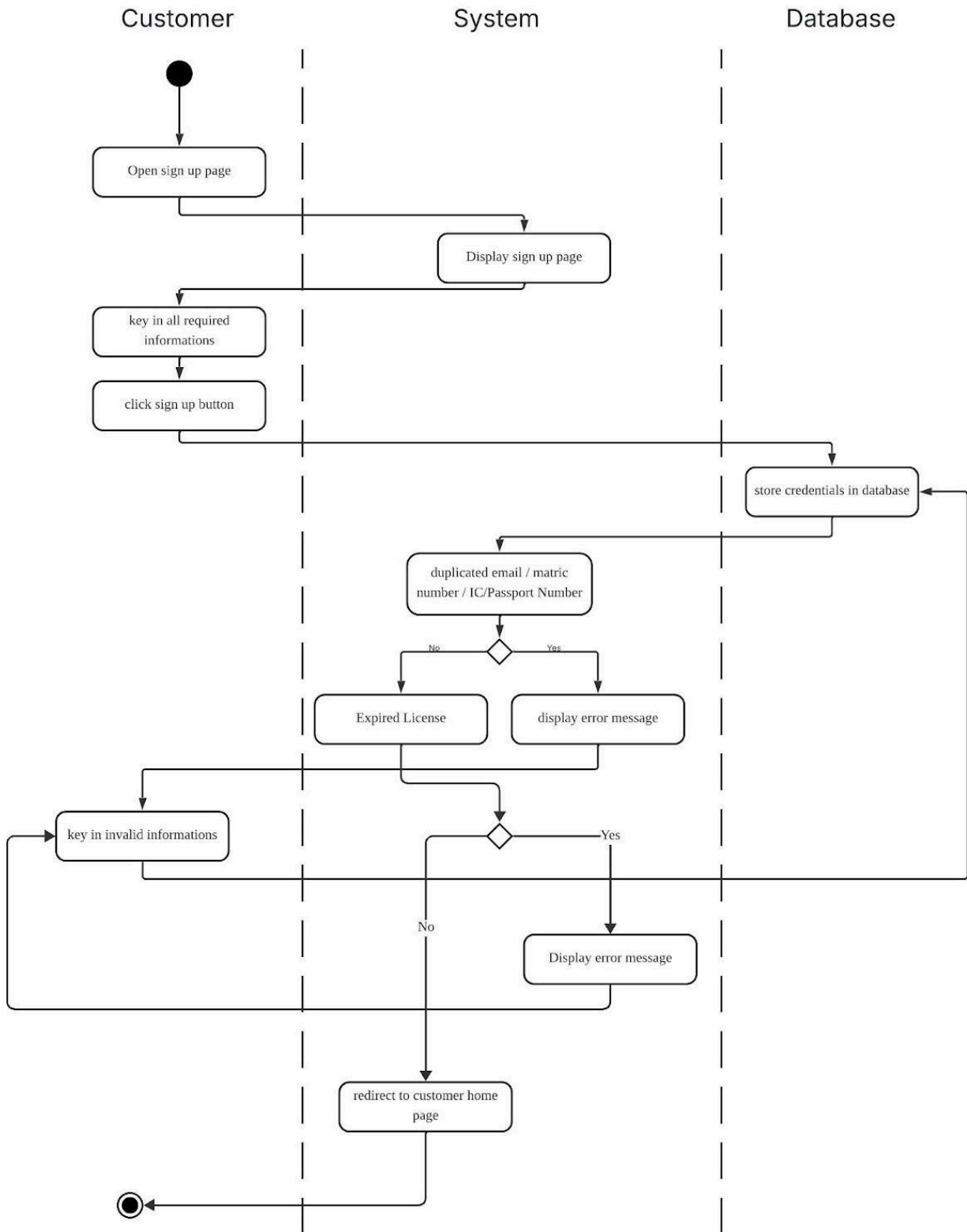


Figure 4.2: Activity Diagram for User Register

4.2.2 UC02: Log In

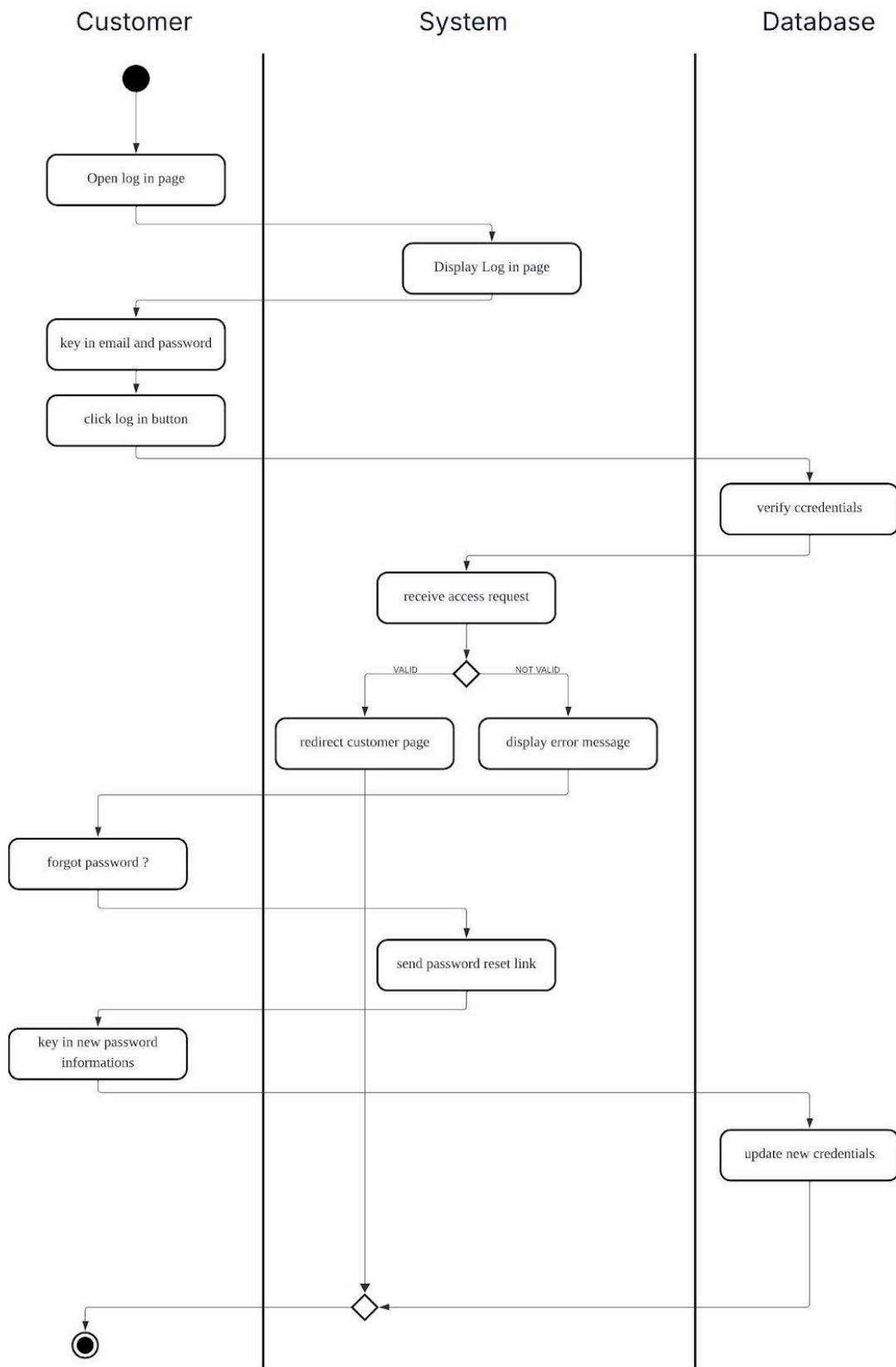


Figure 4.3: Activity Diagram for Log In

4.2.3 UC03: Browse Vehicle

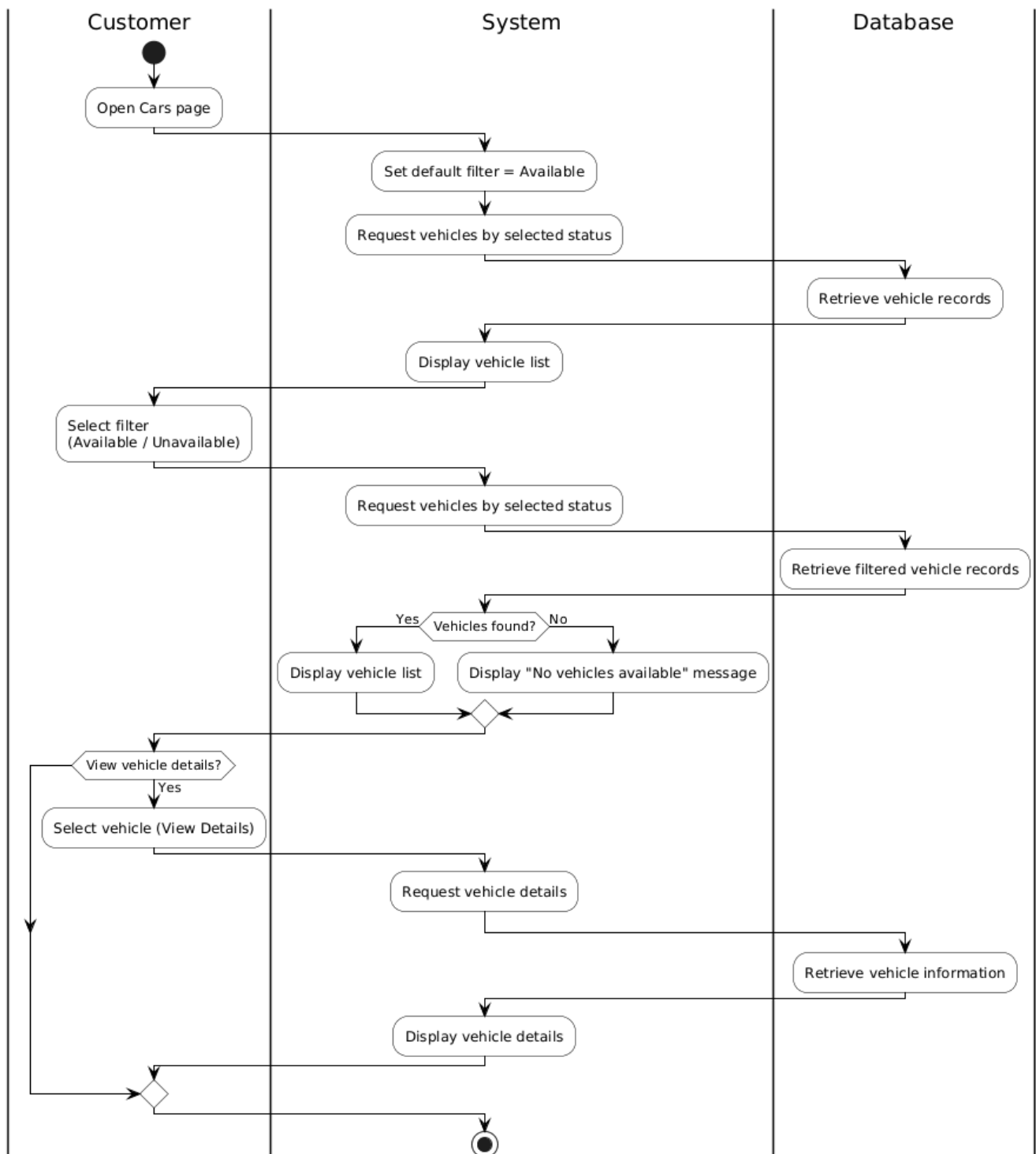


Figure 4.4: Activity Diagram for Browse Vehicle

4.2.4 UC04: Upload Documents

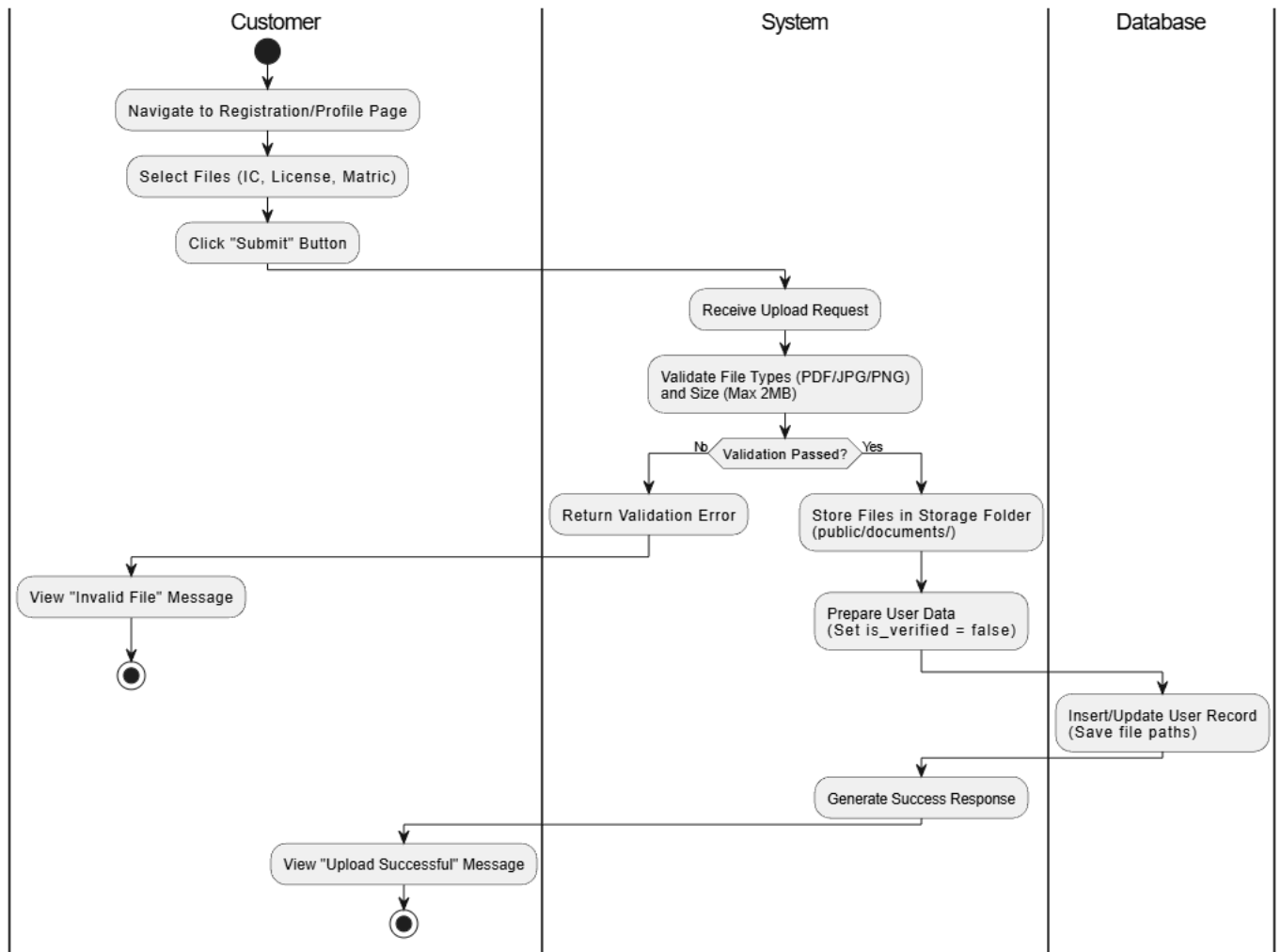


Figure 4.5: Activity Diagram for Upload Documents

4.2.5 UC05: Verify Documents

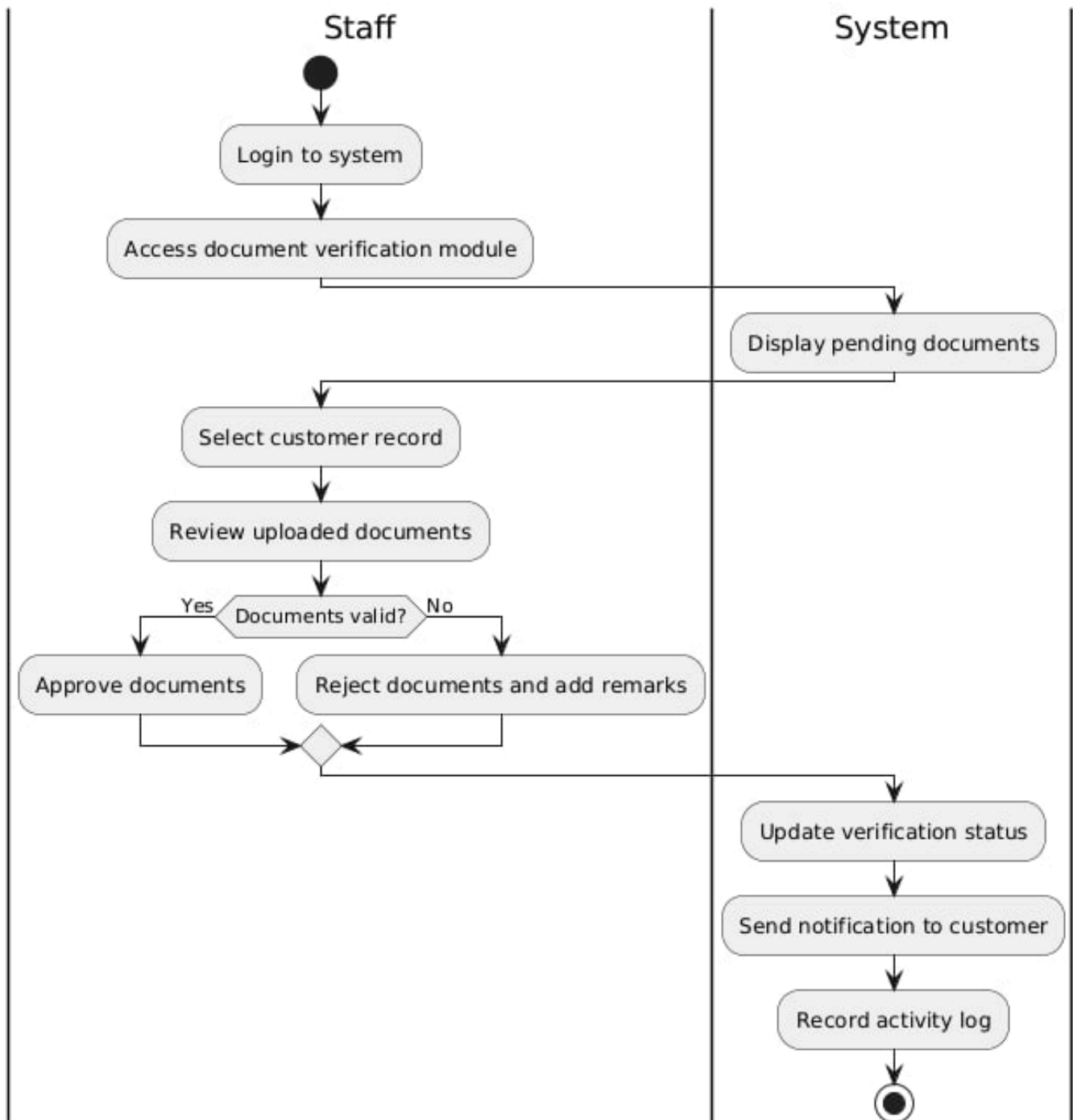


Figure 4.6: Activity Diagram for Verify Documents

4.2.6 UC06: Edit Customer Profile

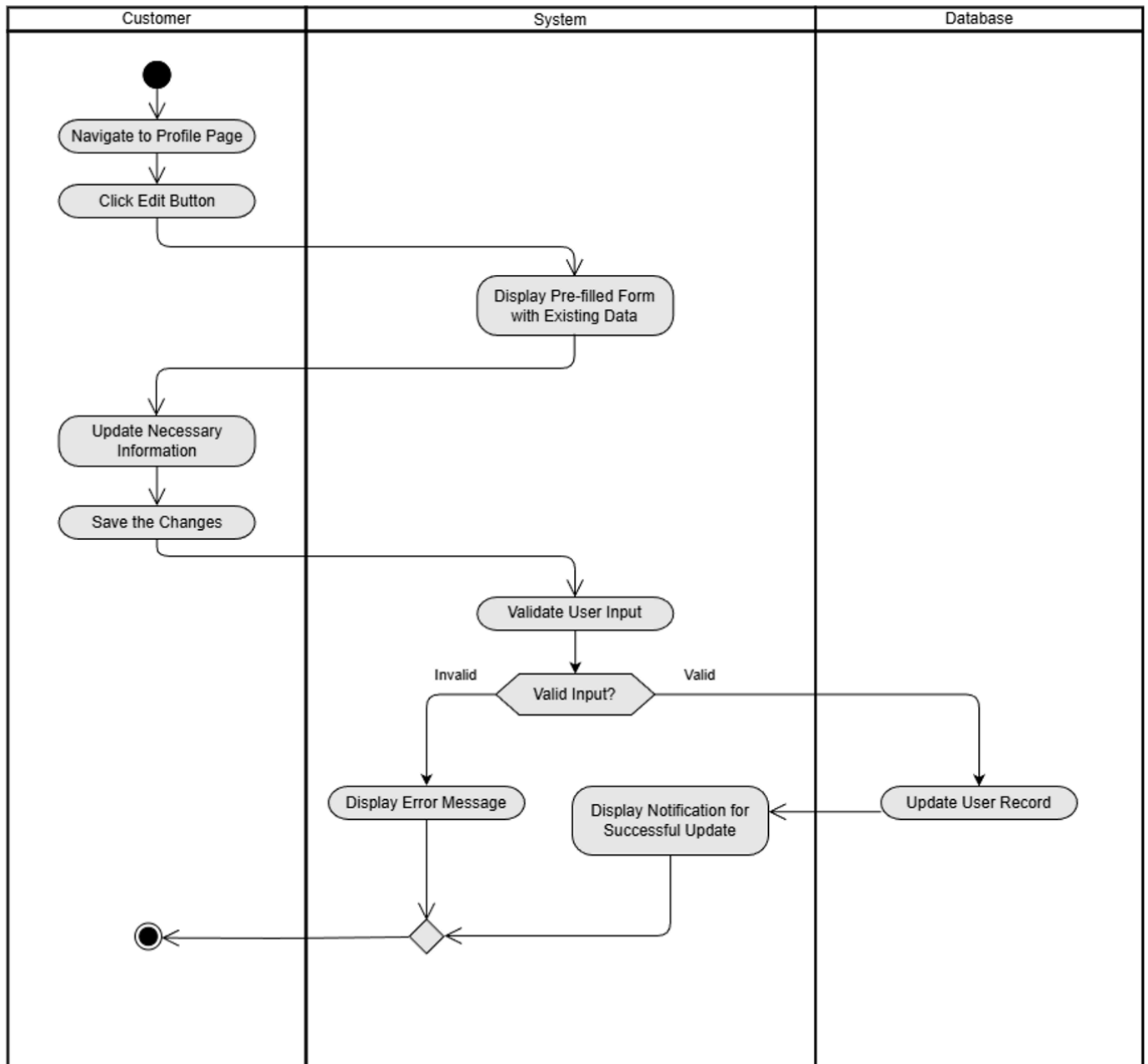


Figure 4.7: Activity Diagram for Edit Customer Profile

4.2.7 UC07: Pickup Vehicle

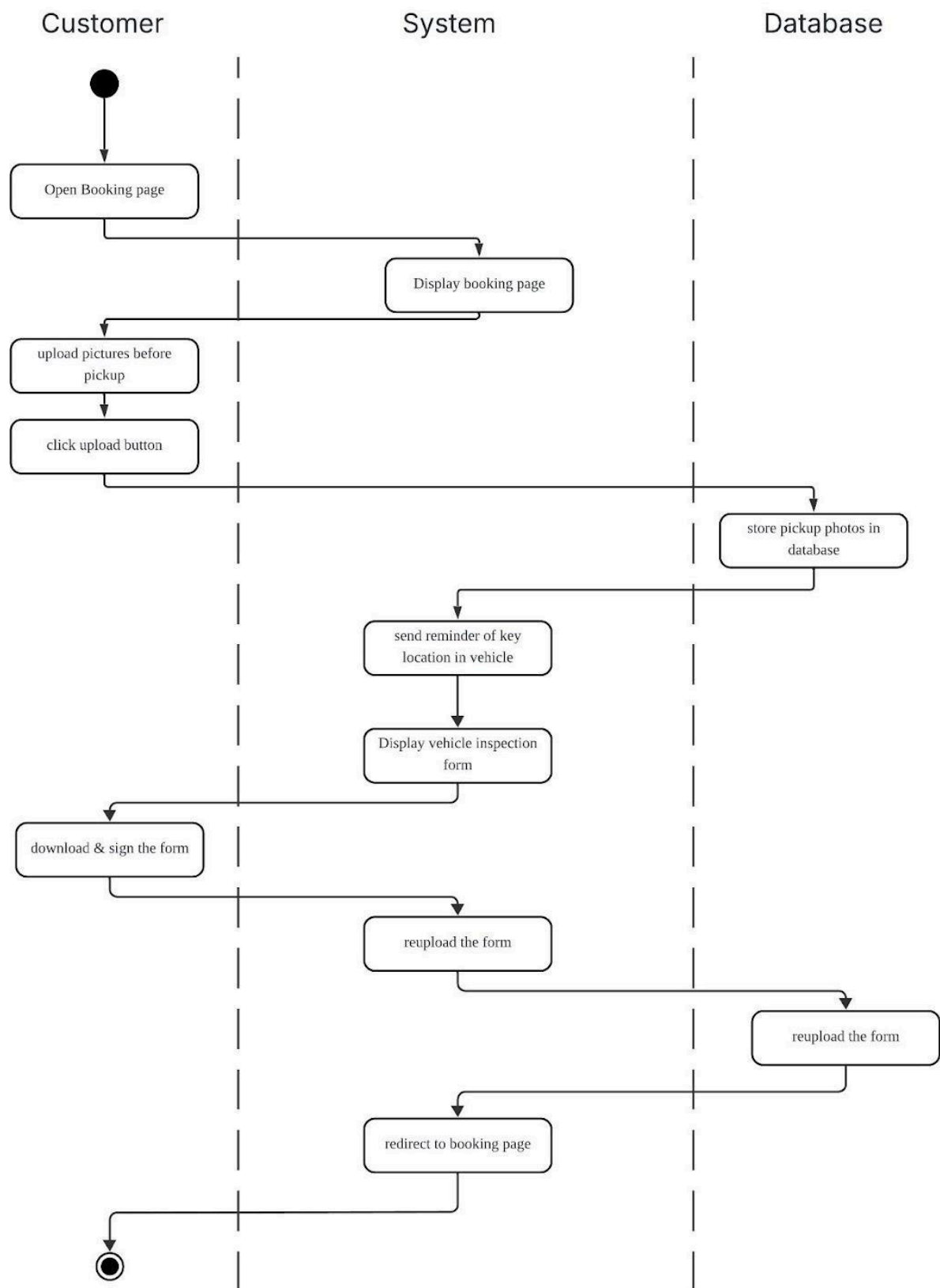


Figure 4.8: Activity Diagram for Pickup Vehicle

4.2.8 UC08: Return Vehicle

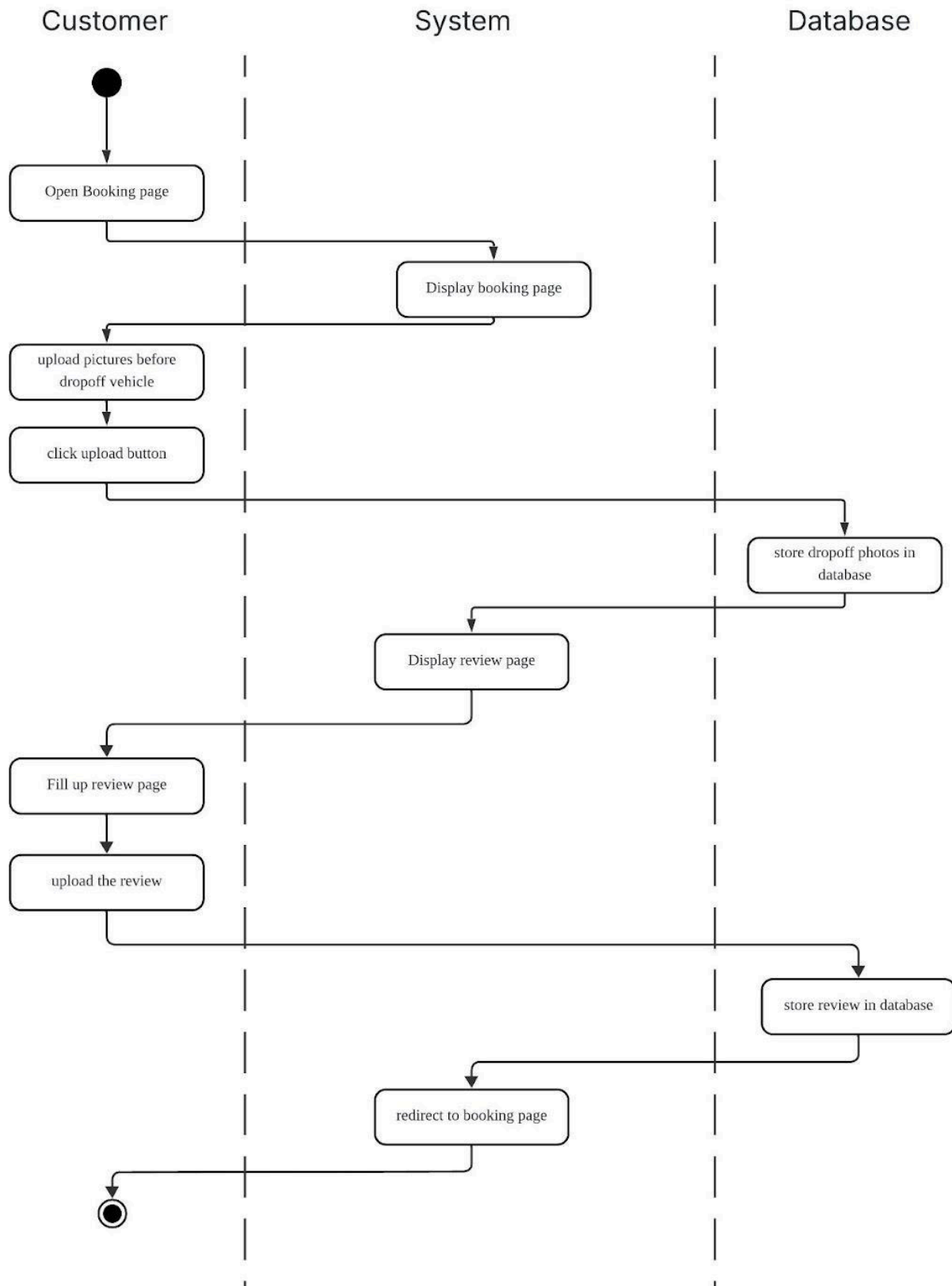


Figure 4.9: Activity Diagram for Return Vehicle

4.2.9 UC09: View Booking History

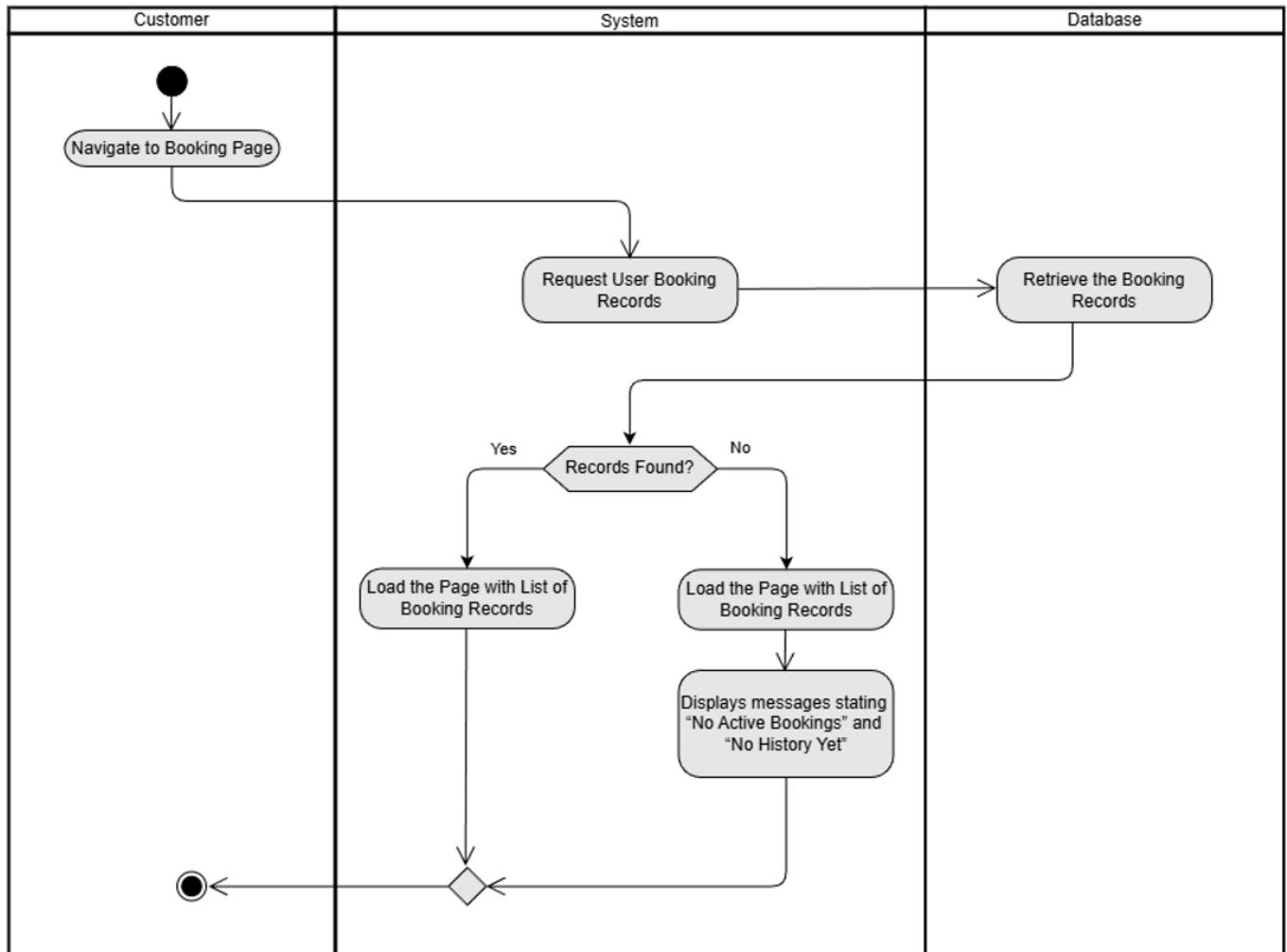


Figure 4.10: Activity Diagram for View Booking History

4.2.10 UC10: Cancel Booking

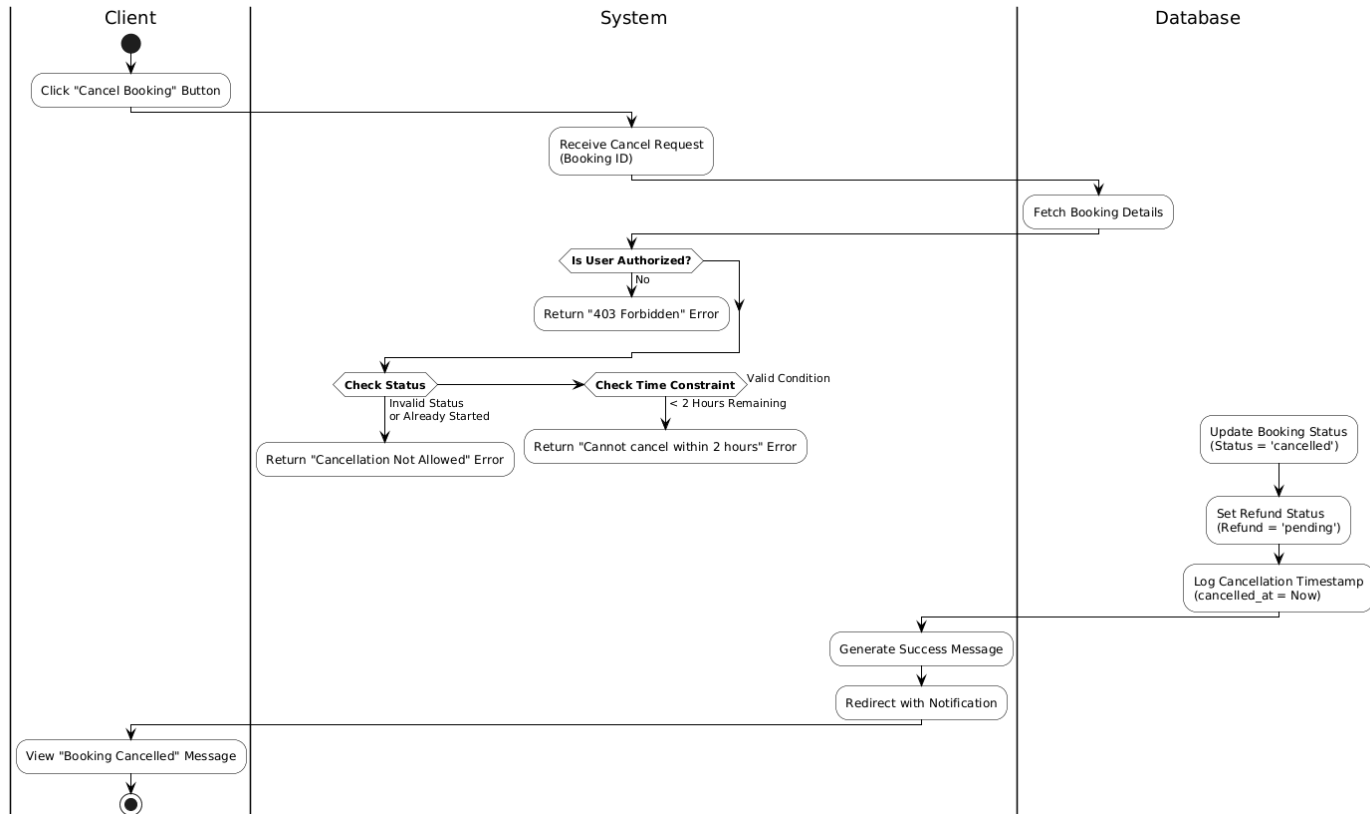


Figure 4.11: Activity Diagram for Cancel Booking

4.2.11 UC11: Booking Vehicle

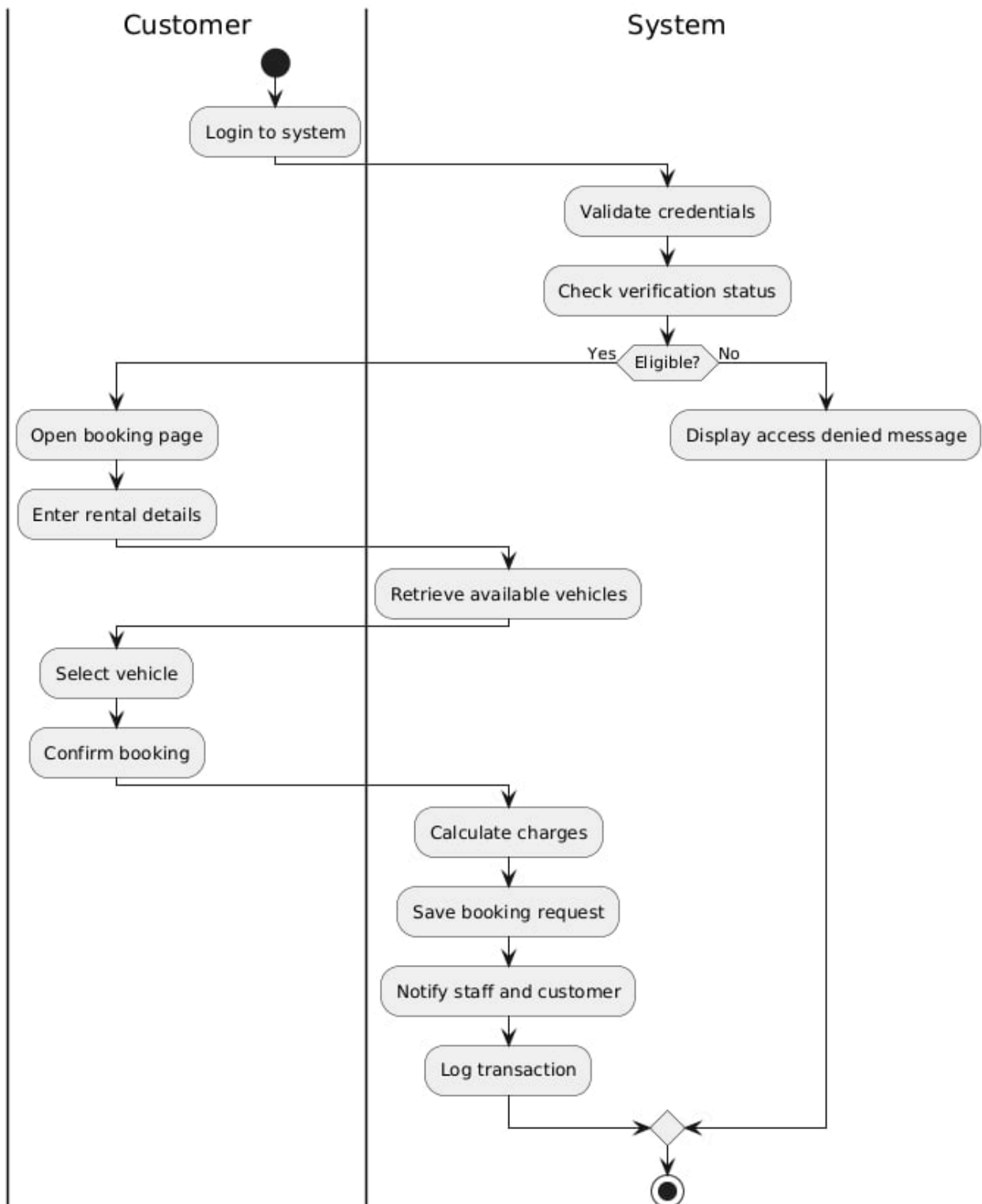


Figure 4.12: Activity Diagram for Booking Vehicle

4.2.12 UC12: Payment

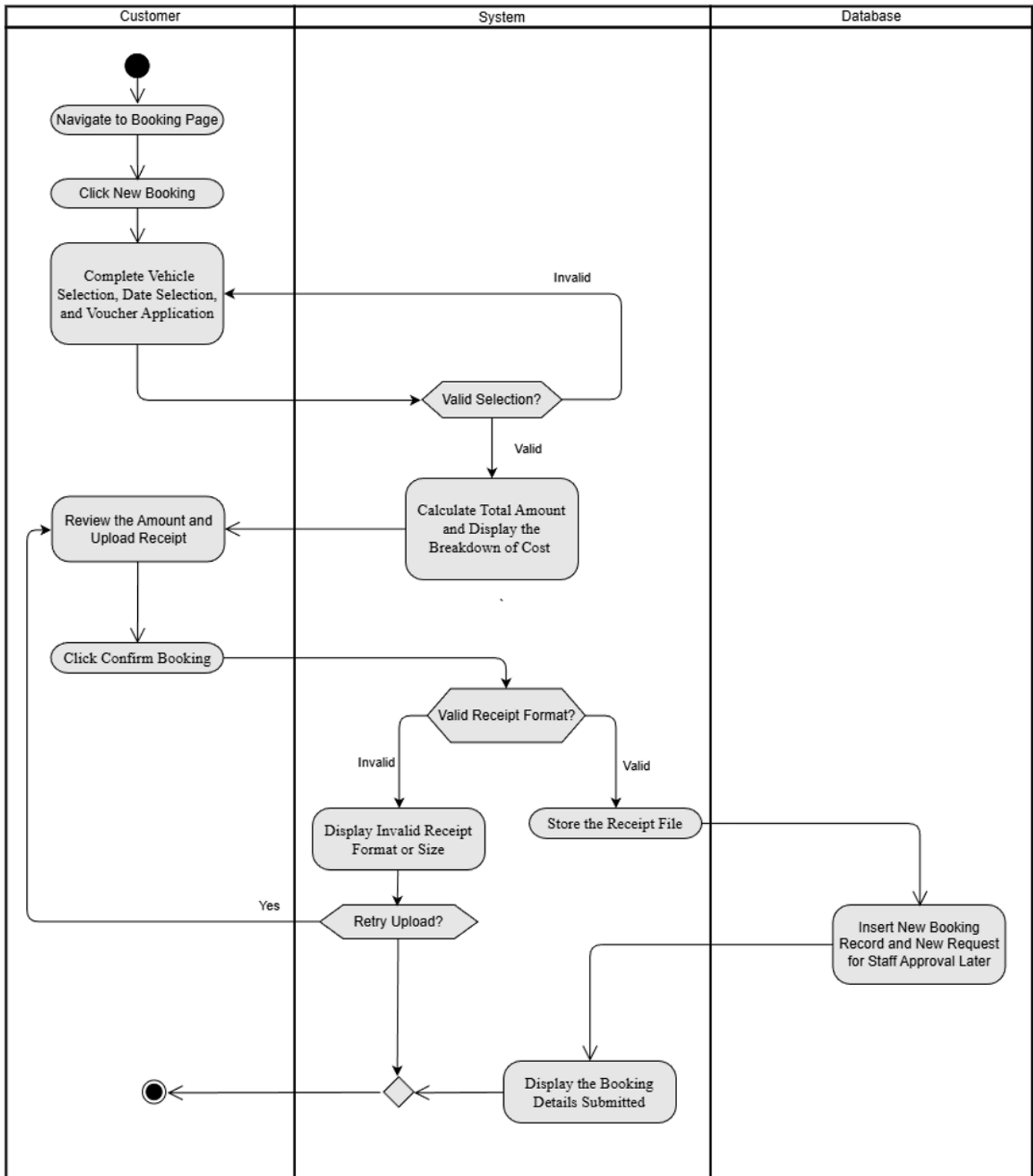


Figure 4.13: Activity Diagram for Payment

4.2.13 UC13: Manage Vehicle

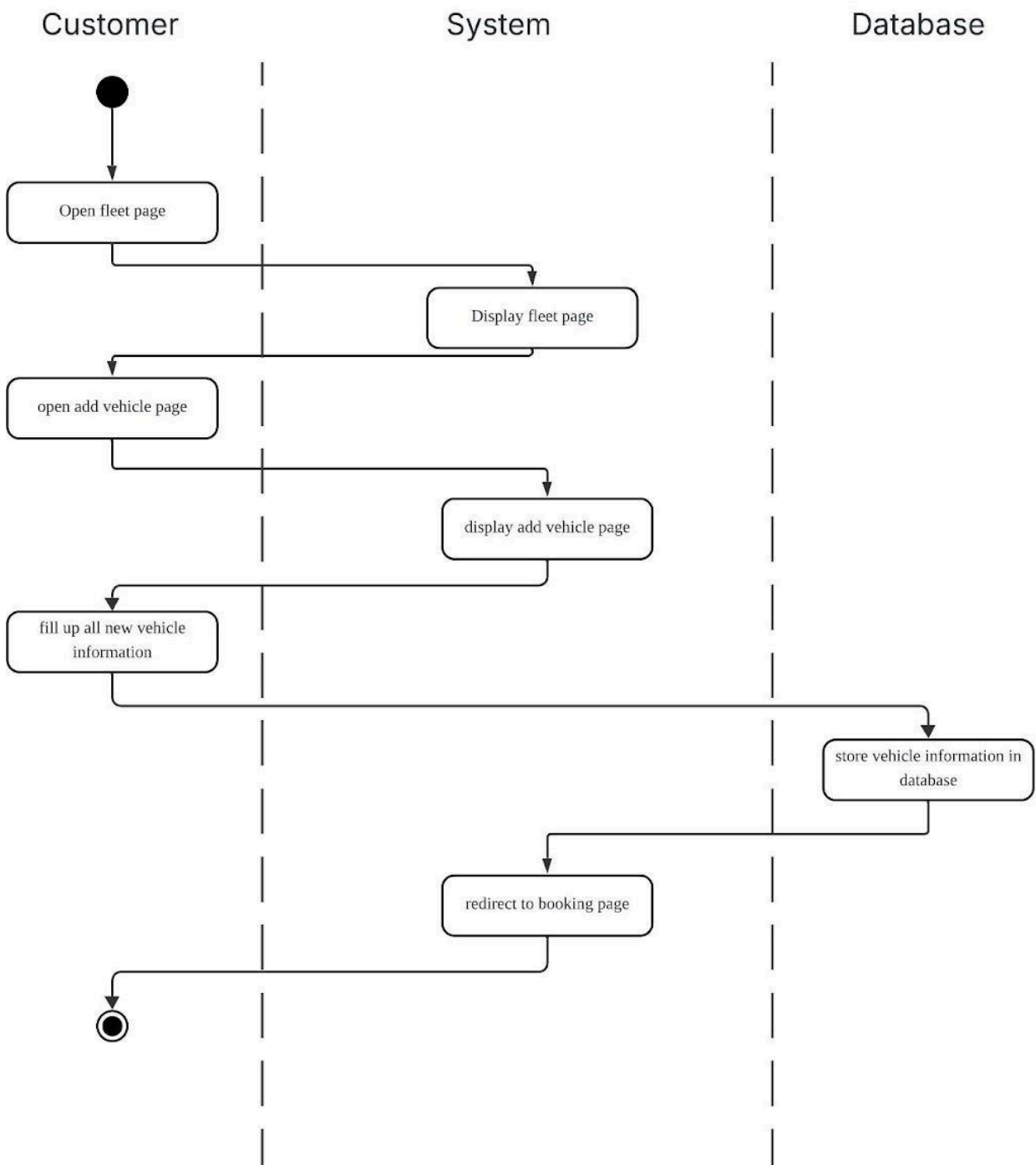


Figure 4.14: Activity Diagram for Manage Vehicle

4.2.14 UC14: Update Vehicle Information

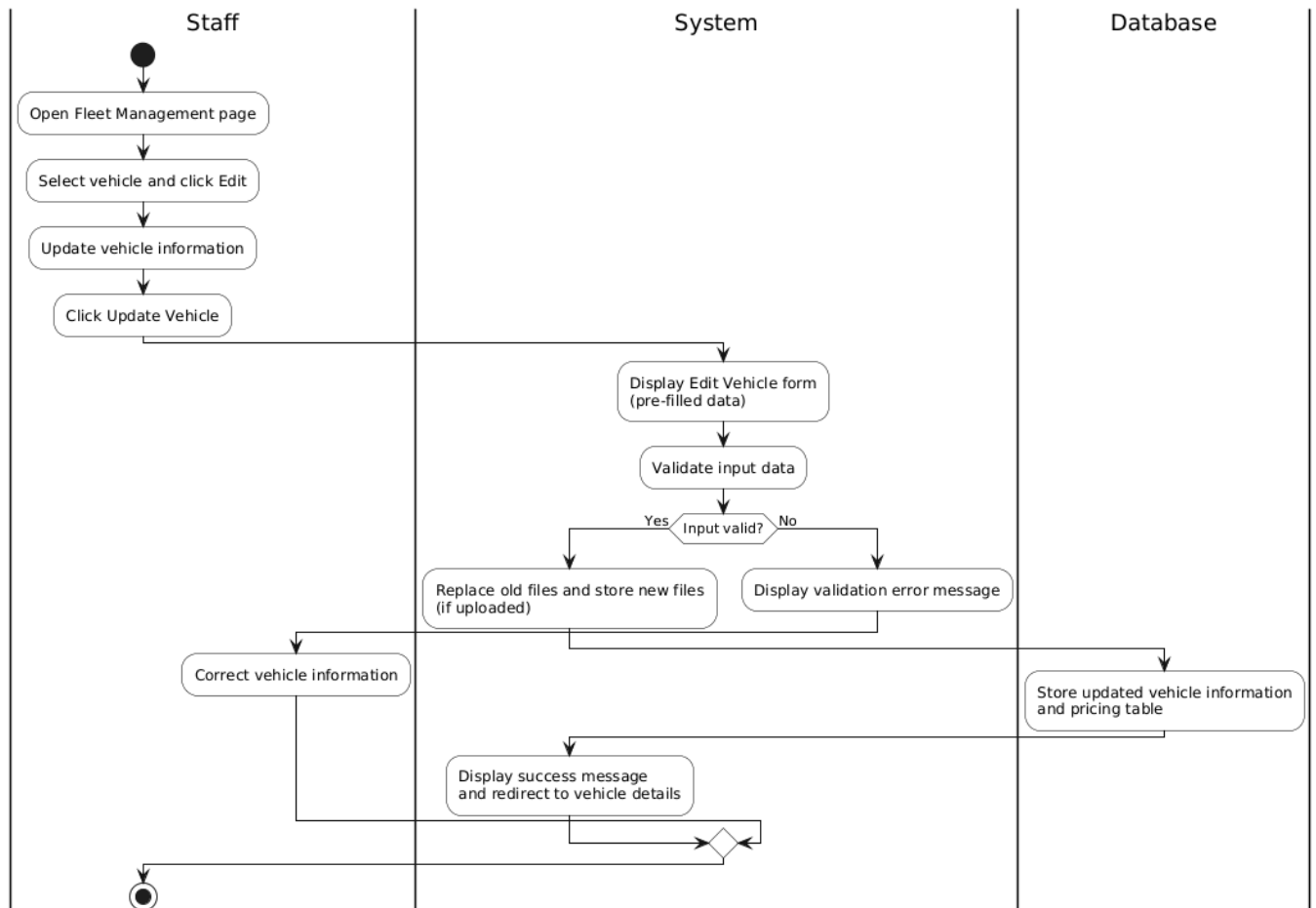


Figure 4.15: Activity Diagram for Update Vehicle Information

4.2.15 UC15: Reporting

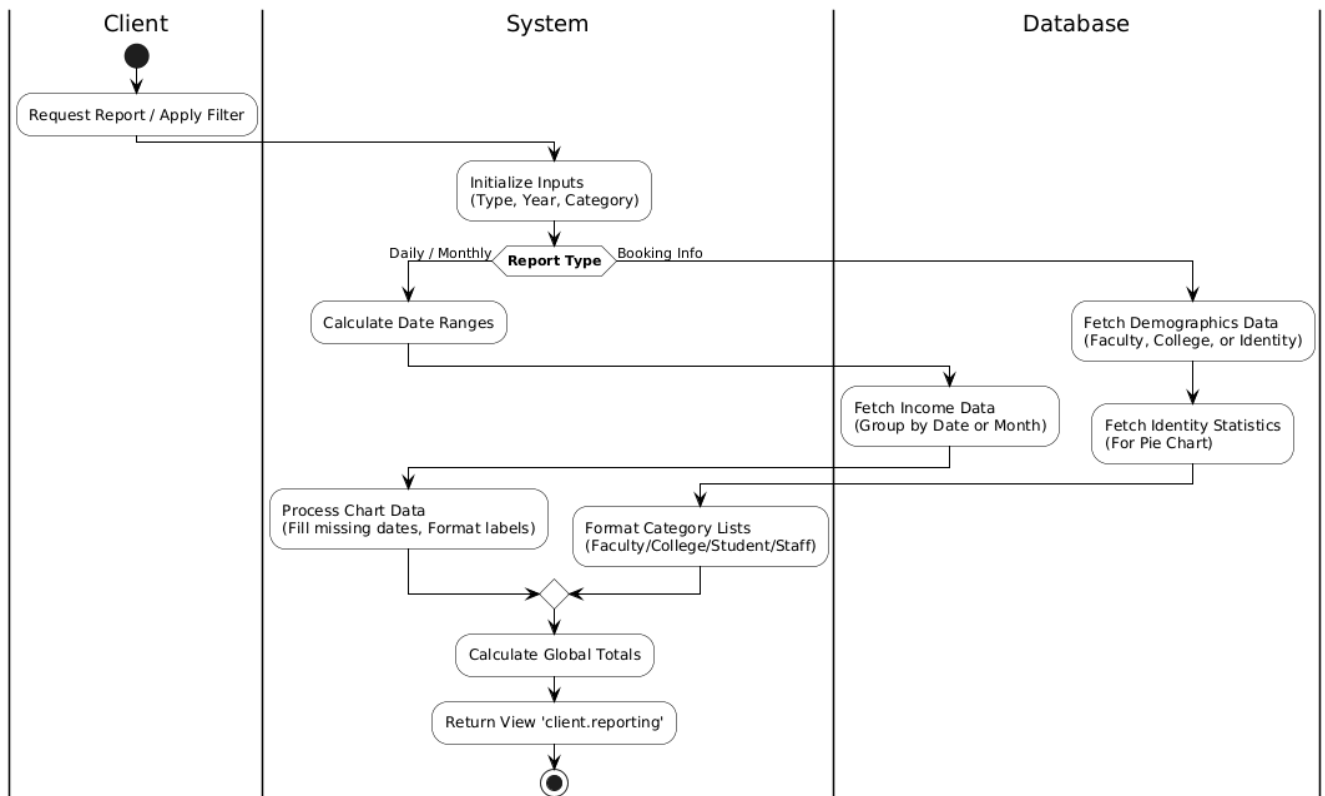


Figure 4.16: Activity Diagram for Reporting

4.2.16 UC16: View All Vouchers

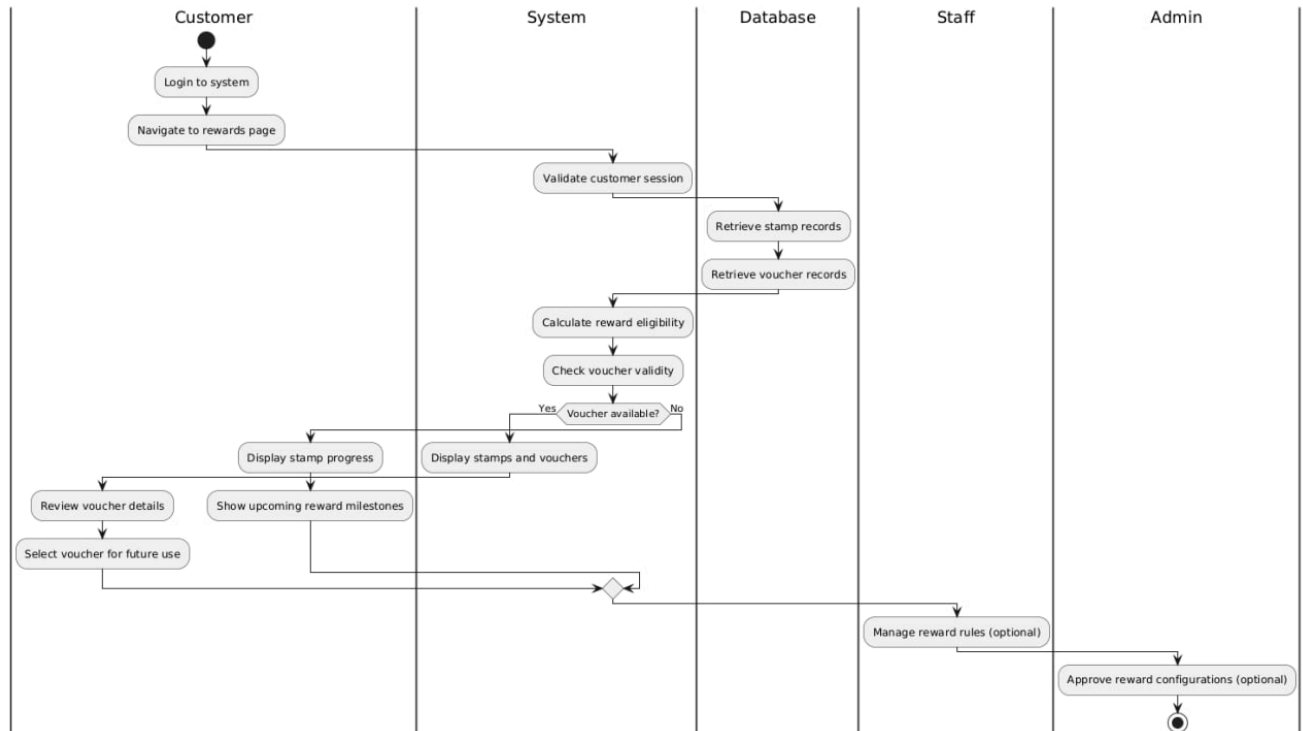


Figure 4.17: Activity Diagram for View All Vouchers

4.3 Meeting Log

Meeting #1

DATE	4 / 11 / 2025 - 10 / 11 / 2025	
LOCATION	Online Meeting (Google Meet)	
FOCUS	Database Planning (ERD)	
AGENDA	1. Initial database structure planning 2. Identify core entities and relationships 3. Draft preliminary ERD	
MEETING MC	HONG JIA BAO	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	4 / 11 / 2025 - 10 / 11 / 2025	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
MINUTES		
Discussion: The group analyzed the system requirements and came up with key entities like User, Booking, Vehicle, Payment, and Reward. The relationships between the entities were also analyzed to make sure that data flow is correct between the modules. Jia Bao was responsible for leading the ERD drafting process and ensuring alignment with system requirements. Decisions Made: - Confirmed core entities for the system: - User - Admin - Staff - Booking - Vehicle - Reward - Agreed to use ERD as the foundation for database design - Relationships and cardinalities were defined. - Identified need for normalization in later stages Action Items: - Jia Bao to prepare first ERD draft - Team members to review and suggest improvements		

Meeting #2

DATE	14 / 11 / 2025	
LOCATION	Student Lounge, Faculty of Computing	
FOCUS	Stakeholder Requirement Fix & Database Revision	
AGENDA	1. Review updated stakeholder feedback 2. Adjust system requirements 3. Revise database structure	
MEETING MC	LEONG JIA LING	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	14 / 11 / 2025	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
GROUP / MEETING PHOTOS		
		

MINUTES

Discussion:

Jia Ling presented new stakeholder requirements gathered from discussions with Hasta Travel. The group examined inconsistencies between earlier requirements and stakeholder expectations. To better represent real-world business procedures, database tables were improved.

Decisions Made:

- Update booking and payment structure
- Improve user verification data storage
- Modify relationships and attributes for better integrity

Action Items:

- Jia Ling to finalize revised requirement list
- Team to update ERD accordingly

Meeting #3

DATE	07 / 12 / 2025	
LOCATION	Student Lounge, Faculty of Computing	
FOCUS	Figma Refinement, Coding Progress & Database Tables	
AGENDA	1. UI/UX prototype refinement 2. Database table structure review 3. Progress check on coding components	
MEETING MC	LEE PEI YUAN	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	07 / 12 / 2025	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
GROUP / MEETING PHOTOS		



MINUTES

Discussion:

Pei Yuan guided the group through improvements in the Figma prototype to match system workflows. Database tables were reviewed to make sure they were compatible with the UI design. To improve reporting and usability, a few table attributes were adjusted.

Decisions Made:

- Align database attributes with UI fields
- Ensure booking flow matches UI logic
- Improve table naming consistency

Action Items:

- Continue UI refinement to match system workflow
- Sync database design with front-end requirements

Meeting #4

DATE	18 / 12 / 2025	
LOCATION	Student Lounge, Faculty of Computing	
FOCUS	Customer Logic & Database Fixes	
AGENDA	1. Review customer booking logic 2. Fix database issues 3. Improve data consistency	
MEETING MC	HENG ZHI QIANG	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	18 / 12 / 2025	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
MINUTES		
Discussion: Noah identified issues in customer booking logic and inconsistencies in database constraints. The group revised validation rules and made sure that the booking conflicts could be prevented. Adjustments were made to customer-related tables. Decisions Made: - Strengthen booking validation logic - Fix foreign key relationships - Improve customer data handling Action Items: - Noah to update booking logic design - Team to re-check database constraints		

Meeting #5

DATE	03 / 01 / 2026	
LOCATION	Student Lounge, Faculty of Computing	
FOCUS	System Testing & Presentation Preparation	
AGENDA	1. System testing using sample data 2. Prepare for presentation 3. Final verification of database design	
MEETING MC	PARTHIV GUNALAN	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	03 / 01 / 2026	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
MINUTES		
Discussion: In order to replicate real-world usage, Parv oversaw end-to-end system testing with realistic data values. The group made sure all modules functioned as it should. Minor issues were found and corrected before final submission and presentation. Decisions Made: - Confirm system readiness for presentation - Validate final database design - Finalize documentation Action Items: - Prepare presentation materials - Conduct final review before submission		